FINITE-DEPTH GRADED BEHAVIOURAL-EQUIVALENCE GAMES ARE SOUND AND COMPLETE

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Editorial scope. Two principal groups only: finite-depth game correspondence; combined infinite-depth game and infinite-term quotient final-coalgebra framework. Direct corollaries, case studies and supporting accessibility facts are retained but not separately counted.

Allowed standard prerequisites (editorial).

- Graded monads and theories, Eilenberg-Moore algebras, coalgebras, kernel pairs/coequalizers and accessible categories. Original definitions, cited standard category-theory facts, and all source-local proofs retained. Standard named prerequisites are not recursively reproved.

1. Introduction

The classical linear-time / branching-time spectrum [20] organizes a plethora of notions of behavioural equivalence on labelled transition systems at various levels of granularity ranging from (strong) bisimilarity to trace equivalence. Similar spectra appear in other system types, e.g. on probabilistic systems, again ranging from branching-time equivalence such as probabilistic bisimilarity to linear-time ones such as probabilistic trace equivalence [27]. While the variation in system types (nondeterministic, probabilistic, etc.) is captured within the framework of *universal coalgebra* [37], the variation in the granularity of equivalence, which we shall generally refer to as the *semantics* of systems, has been tackled, in coalgebraic generality, in a variety of approaches [22, 25, 26, 30]. One setting that manages to accommodate large portions of the linear-time / branching-time spectrum, notably including also intermediate equivalences such as ready similarity, is based on *graded monads* [15, 18, 35].

An important role in the theoretical and algorithmic treatment of a behavioural equivalence is classically played by equivalence games [20, 40], e.g. in partial-order techniques [24] or in on-the-fly equivalence checking [23]. In the present work, we contribute to *graded semantics* in the sense indicated above by showing that, under mild conditions, we can extract from a given graded monad a Spoiler-Duplicator game [40] that characterizes the respective equivalence, i.e. ensures that two states are equivalent under the semantics iff Duplicator wins the game.

As the name suggests, graded monads provide an *algebraic* view on system equivalence; they correspond to *graded theories*, i.e. algebraic theories equipped with a notion of *depth* on their operations. It has been noticed early on [35] that many desirable properties of a semantics depend on this theory being *depth-1*, i.e. having only equations between terms that are uniformly of depth 1. Standard examples include distribution of actions over non-deterministic choice (trace semantics) or monotonicity of actions w.r.t. the choice ordering (similarity) [15]. Put simply, our generic equivalence game plays out an equational proof in a depth-1 equational theory in a somewhat nontraditional manner: Duplicator starts a round by playing a set of equational assumptions she claims to hold at the level of successors of the present state, and Spoiler then challenges one of these assumptions.

In many concrete cases, the game can be rearranged in a straightforward manner to let Spoiler move first as usual; in this view, the equational claims of Duplicator roughly correspond to a short-term strategy determining the responses she commits to playing after Spoiler's next move. In particular, the game instantiates, after such rearrangement, to the standard pebble game for bisimilarity. We analyse additional cases, including similarity and trace equivalence, in more detail. In the latter case, several natural variants of the game arise by suitably restricting strategies played by Duplicator.

It turns out that the game is morally played on a form of pre-determinization of the given coalgebra, which lives in the Eilenberg-Moore category of the zero-th level of the graded monad, and as such generalizes a determinization construction that applies in certain instances of coalgebraic language semantics of automata [26]. Under suitable conditions on the graded monad, this pre-determinization

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indeed functions as an actual determinization, i.e. it turns the graded semantics into standard coalgebraic behavioural equivalence for a functor that we construct on the Eilenberg-Moore category. This construction simultaneously generalizes, for instance, the standard determinization of serial labelled transition systems for trace equivalence and the identification of similarity as behavioural equivalence for a suitable functor on posets [28] (specialized to join semilattices).

While graded semantics has so far been constrained to apply only to finite-depth equivalences (finite-depth bisimilarity, finite trace equivalence, etc.), we obtain, under the mentioned conditions on the graded monad, a new notion of infinite-depth equivalence induced by a graded semantics, namely via the (pre-)determinization. It turns out the natural infinite version of our equivalence game captures precisely this infinite-depth equivalence. This entails a fixpoint characterization of graded semantics on finite systems, giving rise to perspectives for a generic algorithmic treatment.

Related Work. Game characterizations of process equivalences are an established theme in concurrency theory; they tend to be systematic but not generic [14, 20]. Work on games for spectra of quantitative equivalences is positioned similarly [16, 17]. The idea of developing (bi)simulation games in coalgebraic generality goes back to work on branching-time simulations based on relators [9]. There is recent highly general work, conducted in a fibrational setting, on so-called codensity games for various notions of bisimilarity [31]. The emphasis in this work is on generality w.r.t. the measure of bisimilarity, covering, e.g. two-valued equivalences, metrics, pre-orders, and topologies, while, viewed through the lens of spectra of equivalences, the focus remains on branching time. The style of the codensity game is inspired by modal logic, in the spirit of coalgebraic Kantorovich liftings [8, 41]; Spoiler plays predicates thought of as arguments of modalities. Work focused more specifically on games for Kantorovich-style coalgebraic behavioural equivalence and behavioural metrics [32] similarly concentrates on the branching-time case. A related game-theoretic characterization is implicit in work on Λ-(bi)similarity [21], also effectively limited to branching-time. Comonadic game semantics [1, 2, 36] proceeds in the opposite way compared to the mentioned work and ours: It takes existing games as the point of departure, and then aims to develop categorical models.

Graded semantics was developed in a line of work mentioned above [15, 18, 35]. The underlying notion of graded monad stems from algebro-geometric work [39] and was introduced into computer science (in substantially higher generality) in work on the semantics of effects [29]. Our pre-determinization construction relates to work on coalgebras over algebras [7].

Organization. We discuss preliminaries on categories, coalgebras, graded monads, and games in Section 2. We recall the key notions of graded algebra and canonical graded algebra in Section 4, and graded semantics in Section 3. We introduce our pre-determinization construction in Section 5, and finite behavioural equivalence games in Section 6. In Section 7, we consider the infinite version of the game, relating it to behavioural equivalence on the pre-determinization. We finally consider specific cases in detail in Section 8.

2. Preliminaries

We assume basic familiarity with category theory [4]. We will review the necessary background on coalgebra [37], graded monads [35, 39], and the standard bisimilarity game [40].

The category of sets. Unless explicitly mentioned otherwise, we will work in the category **Set** of sets and functions (or *maps*), which is both complete and cocomplete. We fix a terminal object $1 = \{\star\}$ and use $!_X$ (or just $!$ if confusion is unlikely) for the unique map $X \to 1$.

In the subsequent sections, we will mostly draw examples from (slight modifications of) the following (endo-)functors on **Set**. The *powerset functor* $\mathscr{P}$ sends each set X to its set of subsets $\mathscr{P}X$, and acts on a map $f\colon X \to Y$ by taking direct images, i.e. $\mathscr{P}f(S) := f[S]$ for $S \in \mathscr{P}X$. We write $\mathscr{P}_\mathsf{f}$ for the *finitary powerset functor* which sends each set to its set of finite subsets; the action of $\mathscr{P}_\mathsf{f}$ on maps is again given by taking direct images. Similarly, $\mathscr{P}^+$ denotes the non-empty powerset functor ($\mathscr{P}^+(X) = \{Y \in \mathscr{P}(X) \mid Y \neq \emptyset\}$), and $\mathscr{P}^+_\mathsf{f}$ its finitary subfunctor ($\mathscr{P}^+_\mathsf{f}(X) = \{Y \in \mathscr{P}_\mathsf{f}(X) \mid Y \neq \emptyset\}$).

We write $\mathcal{D}X$ for the set of *distributions* on a set X: maps $\mu\colon X \to [0,1]$ such that $\sum_{x\in X} \mu(x) = 1$. A distribution μ is *finitely supported* if the set $\{x \in X \mid \mu(x) \neq 0\}$ is finite. The set of finitely supported distributions on X is denoted $\mathcal{D}_f X$. The assignment $X \mapsto \mathcal{D}X$ is the object-part of a functor: given $f\colon X \to Y$, the map $\mathcal{D}f\colon \mathcal{D}X \to \mathcal{D}Y$ assigns to a distribution $\mu \in \mathcal{D}X$ the *image* distribution $\mathcal{D}f(\mu)\colon Y \to [0,1]$ defined by $\mathcal{D}f(\mu)(y) = \sum_{x\in X | f(x)=y} \mu(x)$. Then, $\mathcal{D}f(\mu)$ is finitely supported if μ is, so $\mathcal{D}_f$ is functorial as well.

Coalgebra. We will review the basic definitions and results of *universal coalgebra* [37], a categorical framework for the uniform treatment of a variety of reactive system types.

Definition 2.1. For an endofunctor $G\colon \mathscr{C} \to \mathscr{C}$ on a category $\mathscr{C}$, a *G-coalgebra* (or just *coalgebra*) is a pair (X, γ) consisting of an object X in $\mathscr{C}$ and a morphism $\gamma\colon X \to GX$. A *(coalgebra) morphism* from (X, γ) to a coalgebra (Y, δ) is a morphism $h\colon X \to Y$ such that $\delta \cdot h = Fh \cdot \gamma$.

Thus, for $\mathscr{C} = \mathbf{Set}$, a coalgebra consists of a set X of *states* and a map $\gamma\colon X \to GX$, which we view as a transition structure that assigns to each state $x \in X$ a structured collection $\gamma(x) \in GX$ of *successors* in X.

Example 2.2. We describe some examples of functors on **Set** and their coalgebras for consideration in the subsequent. Fix a finite set $\mathcal{A}$ of *actions*.

(1) Coalgebras for the functor $G = \mathscr{P}(\mathcal{A} \times -)$ are just *$\mathcal{A}$-labelled transition systems (LTS)*: Given such a coalgebra (X, γ), we can view the elements $(a, y) \in \gamma(x)$ as the a-successors of x. We call (X, γ) *finitely branching* (resp. *serial* if $\gamma(x)$ is finite (resp. non-empty) for all $x \in X$. Finitely branching (resp. serial) LTS are coalgebras for the functor $G = \mathscr{P}_{\mathsf{f}}(\mathcal{A} \times -)$ (resp. $\mathscr{P}^+(\mathcal{A} \times -)$).

(2) A coalgebra (X, γ) for the functor $G = \mathcal{D}(\mathcal{A} \times -)$ is a *(generative) probabilistic transition system* (PTS): The transition structure γ assigns to each state $x \in X$ a distribution $\gamma(x)$ on pairs $(a, y) \in \mathcal{A} \times X$. We think of $\gamma(x)(a, y)$ as the probability of executing an a-transition to state y while sitting in state x. A PTS (X, γ) is *finitely branching* if $\gamma(x)$ is finitely supported for all $x \in X$; then, finitely branching PTS are coalgebras for $\mathcal{D}_f(\mathcal{A} \times -)$.

Given coalgebras (X, γ) and (Y, δ) for an endofunctor G on **Set**, states $x \in X$ and $y \in Y$ are *G-behaviourally equivalent* if there exist coalgebra morphisms

$$(X, \gamma) \xrightarrow{f} (Z, \zeta) \xleftarrow{g} (Y, \delta)$$

such that $f(x) = g(y)$. Behavioural equivalence can be approximated via the (initial ω-segment of the) *final chain* $(G^n 1)_{n \in \omega}$, where G^n denotes n-fold application of G. The *canonical cone* of a coalgebra (X, γ) is then the family of maps $\gamma_n\colon X \to G^n 1$ defined inductively for $n \in \omega$ by

$$\gamma_0 = \big(X \xrightarrow{!} 1\big), \text{and}$$
$$\gamma_{n+1} = \big(X \xrightarrow{\gamma} GX \xrightarrow{G\gamma_n} GG^n 1 = G^{n+1} 1\big).$$

States $x, y \in X$ are *finite-depth behaviourally equivalent* if $\gamma_n(x) = \gamma_n(y)$ for all $n \in \omega$.

Remark 2.3. It follows from results of Worrell [42] that behavioural equivalence and finite-depth behavioural equivalence coincide for finitary functors on **Set**, where a functor G on **Set** is *finitary* if it preserves filtered colimits. Equivalently, for every set X and each $x \in GX$ there exists a finite subset $Y \subseteq X$ such that $x = Gi[GY]$, where $i\colon Y \hookrightarrow X$ is the inclusion map [5, Cor. 3.3].

Bisimilarity games. We briefly recapitulate the classical *bisimilarity game*, a two-player graph game between the players Duplicator (D) and Spoiler (S); player D tries to show that two given states are bisimilar, while S tries to refute this. *Configurations* of the game are pairs $(x, y) \in X \times X$ of states in a LTS (X, γ). The game proceeds in rounds, starting from the *initial configuration*, which is just the contested pair of states. In each round, starting from a configuration (x, y), S picks one of the sides, say, x, and then selects an action $a \in \mathcal{A}$ and an a-successor x' of x; player D then selects a corresponding successor on the other side, in this case an a-successor y' of y. The game then reaches the new configuration (x', y'). If a player gets stuck, the play is *winning* for their opponent, whereas any infinite play is winning for D.

It is well known (e.g. [40]) that D has a winning strategy in the bisimilarity game at a configuration (x, y) iff (x, y) is a pair of bisimilar states. Moreover, for finitely branching LTS, an equivalent formulation may be given in terms of the *n-round bisimilarity game*: the rules of the n-round game are the same as those above, only now D wins as soon as at most n rounds have been played. In fact, a configuration (x, y) is a bisimilar pair precisely if D has a winning strategy in the n-round bisimilarity game for all $n \in \omega$.

We mention just one obvious variation of this game that characterizes a different spot on the linear-time/branching-time spectrum: The *mutual-simulation game* is set up just like the bisimulation game, except that S may only choose his side once, in the first round, and then has to move on that side

in all subsequent rounds (in the bisimulation game, he can switch sides in every round if he desires). It is easily checked that states x, y are mutually similar iff S wins the position (x, y) in the mutual-simulation game. We will see that both these games (and many others) are obtained as instances of our generic notion of graded equivalence game.

Graded monads. We now review some background material on graded monads [35, 39]:

Definition 2.4. A *graded monad* $\mathbb{M}$ on a category $\mathscr{C}$ is a triple (M, η, μ) where M is a family of functors $M_n \colon \mathscr{C} \to \mathscr{C}$ on $\mathscr{C}$ $(n \in \omega)$, $\eta \colon \mathsf{id} \to M_0$ is a natural transformation (the *unit*), and μ is a family of natural transformations

$$(n, k \in \omega) \qquad \mu^{n,k} \colon M_n M_k \to M_{n+k}$$

(the *multiplication*) such that the following diagrams commute for all $n, m, k \in \omega$:

(2.1)
$$\begin{array}{ccccc} & & M_n & & \\ & \swarrow^{M_n\eta} & \downarrow{\scriptstyle \mathsf{Id}} & \searrow^{\eta M_n} & \\ M_nM_0 & \xrightarrow{\mu^{n,0}} & M_n & \xleftarrow{\mu^{0,n}} & M_0M_n \end{array}$$

(2.2)
$$\begin{array}{ccc} M_nM_kM_m & \xrightarrow{M_n\mu^{k,m}} & M_nM_{k+m} \\ {\scriptstyle \mu^{n,k}M_m}\downarrow & & \downarrow{\scriptstyle \mu^{n,k+m}} \\ M_{n+k}M_m & \xrightarrow{\mu^{n+k,m}} & M_{n+k+m} \end{array}$$

We refer to (2.1) and (2.2) as the *unit* and *associative* laws of $\mathbb{M}$, respectively. We call $\mathbb{M}$ *finitary* if all of the functors $M_n \colon \mathscr{C} \to \mathscr{C}$ are finitary.

The above notion of graded monad is due to Smirnov [39]. Katsumata [29], Fujii et al. [19], and Mellies [34] consider a more general notion of graded (or *parametrized*) monad given as a lax monoidal action of a monoidal category $\mathscr{M}$ (representing the system of grades) on a category $\mathscr{C}$. Graded monads in the above sense are recovered by taking $\mathscr{M}$ to be the (discrete category induced by the) monoid $(\mathbb{N}, +, 0)$.

The graded monad laws imply that the triple $(M_0, \eta, \mu^{0,0})$ is a (plain) monad on the base category $\mathscr{C}$; we use this freely without further mention.

Example 2.5. We review some salient constructions [35] of graded monads on **Set** for later use.

(1) Every endofunctor G on **Set** induces a graded monad $\mathbb{M}_G$ with underlying endofunctors $M_n := G^n$ (the n-fold composite of G with itself); the unit $\eta_X \colon X \to G^0X = X$ and multiplication $\mu^{n,k}_X \colon G^nG^kX \to G^{n+k}X$ are all identity maps. We will later see that $\mathbb{M}_G$ captures (finite-depth) G-behavioural equivalence.

(2) Let (T, η, μ) be a monad on **Set**, let F be an endofunctor on **Set**, and let $\lambda \colon FT \to TF$ be a natural transformation such that

$$\lambda \cdot F\eta = \eta F \qquad \text{and} \qquad \lambda \cdot F\mu = \mu F \cdot T\lambda \cdot \lambda T$$

(i.e. λ is a distributive law of the functor F over the monad T). For each $n \in \omega$, let $\lambda^n \colon F^nT \to TF^n$ denote the natural transformation defined inductively by

$$\lambda^0 := \mathsf{id}_T; \qquad \lambda^{n+1} := \lambda^n F \cdot F^n\lambda.$$

We obtain a graded monad with $M_n := TF^n$, unit η, and components $\mu^{n,k}$ of the multiplication given as the composites

$$TF^nTF^k \xrightarrow{T\lambda^nF^k} TTF^nF^k = TTF^{n+k} \xrightarrow{\mu F^{n+k}} TF^{n+k}.$$

Such graded monads relate strongly to Kleisli-style coalgebraic trace semantics [22].

(3) We obtain (by instance of the example above) a graded monad $\mathbb{M}_T(\mathcal{A})$ with $M_n = T(\mathcal{A}^n \times -)$ for every monad T on **Set** and every set $\mathcal{A}$. Thus, $\mathbb{M}_T$ is a graded monad for traces under effects specified by T; e.g. for $T = \mathcal{D}$, we will see that $\mathbb{M}_T(\mathcal{A})$ captures probabilistic trace equivalence on PTS.

(4) Similarly, given a monad T, an endofunctor F, both on the same category $\mathscr{C}$, and a distributive law $\lambda\colon TF \to FT$ of T over F, we obtain a graded monad with $M_n := F^nT$, unit and multiplication given analogously as in item (2) above (see [35, Ex. 5.2.6]). Such graded monads relate strongly to Eilenberg-Moore-style coalgebraic language semantics [12].

Graded variants of Kleisli triples have been introduced and proved equivalent to graded monads (in a more general setting) by Katsumata [29]:

Notation 2.6. We will employ the *graded Kleisli star* notation: for $n \in \omega$ and a morphism $f\colon X \to M_kY$, we write

$$f_n^* := \big(M_nX \xrightarrow{M_nf} M_nM_k \xrightarrow{\mu^{n,k}} M_{n+k}Y\big). \tag{2.3}$$

In this way, we obtain a morphism satisfying the following graded variants [29, Def. 2.3] of the usual laws of the Kleisli star operation for ordinary monads: for every $m \in \omega$ and morphisms $f\colon X \to M_nY$ and $g\colon Y \to M_kZ$ we have:

$$f_0^* \cdot \eta_X = f, \tag{2.4}$$

$$(\eta_X)_n^* = \mathsf{id}_{M_nX}, \tag{2.5}$$

$$(g_n^* \cdot f)_m^* = g_{m+n}^* \cdot f_m^*. \tag{2.6}$$

Graded theories. Graded theories, in a generalized form in which arities of operations are not restricted to be finite, have been proved equivalent to graded monads on **Set** [35] (the finitary case was implicitly covered already by Smirnov [39]). We work primarily with the finitary theories below; we consider infinitary variants of such theories only when considering infinite-depth equivalences (Section 7).

Definition 2.7. (1) A *graded signature* is a set Σ of *operations* f equipped with a finite *arity* $\mathsf{ar}(f) \in \omega$ and a finite *depth* $d(f) \in \omega$. An operation of arity 0 is called a *constant*.

(2) Let X be a set of *variables* and let $n \in \omega$. The set $\mathscr{T}_{\Sigma,n}(X)$ of Σ-*terms of uniform depth* n *with variables in* X is defined inductively as follows: every variable $x \in X$ is a term of uniform depth 0 and, for $f \in \Sigma$ and $t_1, \dots, t_{\mathsf{ar}(f)} \in \mathscr{T}_{\Sigma,k}(X)$, $f(t_1, \dots, t_{\mathsf{ar}(f)})$ is a Σ-term of uniform depth $k + d(f)$. In particular, a constant c has uniform depth k for all $k \geq d(c)$.

(3) A *graded* Σ-*theory* is a set $\mathcal{E}$ of *uniform-depth equations*: pairs (s,t), written '$s = t$', such that $s, t \in \mathscr{T}_{\Sigma,n}(X)$ for some $n \in \omega$; we say that (s,t) is *depth-n*. A theory is *depth-n* if all of its equations and operations have depth at most n.

Notation 2.8. A *uniform-depth substitution* is a map $\sigma\colon X \to \mathscr{T}_{\Sigma,k}(Y)$, where $k \in \omega$ and X, Y are sets. Then σ extends to a family of maps $\bar{\sigma}_n\colon \mathscr{T}_{\Sigma,n}(X) \to \mathscr{T}_{\Sigma,k+n}(Y)$ $(n \in \omega)$ defined recursively by

$$\bar{\sigma}_n(f(t_1, \dots, t_{\mathsf{ar}(f)})) = f(\bar{\sigma}_m(t_1), \dots, \bar{\sigma}_m(t_{\mathsf{ar}(f)})),$$

where $t_i \in \mathscr{T}_{\Sigma,m}$ and $d(f) + m = n$. For a term $t \in T_{\Sigma,k}(X)$, we also write $t\sigma := \bar{\sigma}_n(t)$ when confusion is unlikely.

Given a graded theory $\mathcal{T} = (\Sigma, \mathcal{E})$, we have essentially the standard notion of equational derivation (sound and complete over graded algebras, cf. Section 4), restricted to uniform-depth equations. Specifically, the system includes the expected rules for reflexivity, symmetry, transitivity, and congruence, and moreover allows substituted introduction of axioms: If $s = t$ is in $\mathcal{E}$ and σ is a uniform-depth substitution, then derive the (uniform-depth) equation $s\sigma = t\sigma$. (A substitution rule that more generally allows uniform-depth substitution into derived equations is then admissible.) For a set Z of uniform-depth equations, we write

$$Z \vdash s = t$$

if the uniform-depth equation $s = t$ is derivable from equations in Z in this system; note that unlike the equational axioms in $\mathcal{E}$, the equations in Z cannot be substituted into in such a derivation (they constitute assumptions on the variables occurring in s, t).

We then see that $\mathcal{T}$ induces a graded monad $\mathbb{M}_\mathcal{T}$ with M_nX being the quotient of $\mathscr{T}_{\Sigma,n}(X)$ modulo derivable equality under $\mathcal{E}$; the unit and multiplication of $\mathbb{M}_\mathcal{T}$ are given by the inclusion of variables as depth-0 terms and the collapsing of layered terms, respectively. Conversely, every graded monad arises from a graded theory in this way [35].

We will restrict attention to graded monads presented by depth-1 graded theories:

Definition 2.9. A *presentation* of a graded monad $\mathbb{M}$ is a graded theory $\mathcal{T}$ such that $\mathbb{M} \cong \mathbb{M}_{\mathcal{T}}$, in the above notation. A graded monad is *depth-1* if it has a depth-1 presentation.

Example 2.10. Fix a set $\mathcal{A}$ of actions. We describe depth-1 graded theories associated (via the induced behavioural equivalence, Section 3) to standard process equivalences on LTS and PTS [15].

(1) The graded theory $\mathsf{JSL}(\mathcal{A})$ of *$\mathcal{A}$-labelled join semilattices* has as depth-1 operations all formal sums

$$\textstyle\sum_{i=1}^{n} a_i(-), \quad \text{for } n \geq 0 \text{ and } a_1, \dots, a_n \in \mathcal{A}$$

(and no depth-0 operations); we write 0 for the empty formal sum. The axioms of $\mathsf{JSL}(\mathcal{A})$ consist of all depth-1 equations $\sum_{i=1}^{n} a_i(x_i) = \sum_{j=1}^{m} b_j(y_j)$ (where the x_i and y_j are variables, not necessarily distinct) such that $\{(a_i, x_i) \mid 1 \leq i \leq n\} = \{(b_j, y_j) \mid 1 \leq j \leq m\}$. The graded monad induced by $\mathsf{JSL}(\mathcal{A})$ is $\mathbb{M}_G$ for $G = \mathscr{P}_{\mathsf{f}}(\mathcal{A} \times (-))$ (cf. Example 2.5.(1)).

(2) The *graded theory of probabilistic traces*, $\mathsf{PT}(\mathcal{A})$, has a depth-0 convex sum operation

$$\textstyle\sum_{i=1}^{n} p_i \cdot (-) \quad \text{for all } p_1, \dots, p_n \in [0,1] \text{ such that } \sum_{i=1}^{n} p_i = 1$$

and unary depth-1 operations $a(-)$ for all actions $a \in \mathcal{A}$. As depth-0 equations, we take the usual equational axiomatisation of convex algebras, which is given by the equation $\sum_{i=1}^{n} \delta_{ij} \cdot x_j = x_i$ (where δ_{ij} denotes the Kronecker delta function) and all instances of the equation scheme

$$\sum_{i=n}^{n} p_i \cdot \sum_{j=1}^{m} q_{ij} \cdot x_j = \sum_{j=1}^{m} \Big(\sum_{i=1}^{n} p_i q_{ij} \Big) \cdot x_j.$$

We further impose depth-1 equations stating that actions distribute over convex sums:

$$a\Big(\sum_{i=1}^{n} p_i \cdot x_i \Big) = \sum_{i=1}^{n} p_i \cdot a(x_i).$$

The theory $\mathsf{PT}(\mathcal{A})$ presents $\mathbb{M}_{\mathcal{D}_f}(\mathcal{A})$, where $\mathcal{D}_f$ is the finitely supported distribution monad (cf. Example 2.5.(3)).

(3) We mention two variations on the graded theory above. First, the *graded theory of (non-deterministic) traces* presenting $\mathbb{M}_{\mathscr{P}_{\mathsf{f}}}(\mathcal{A})$ has depth-0 operations $+, 0$ and equations for join-semilattices with bottom, and unary depth-1 operations a for $a \in \mathcal{A}$ as in (1) above; the depth-1 equations now state that actions distribute over joins and preserve bottom. Second, the *graded theory of serial (non-deterministic) traces* arises by omitting 0 and associated axioms from the graded theory of traces, and yields a presentation of $\mathbb{M}_{\mathscr{P}^{+}_{\mathsf{f}}}(\mathcal{A})$.

(4) The *graded theory of simulation* has the same signature and depth-0 equations as the graded theory of traces, along with depth-1 equations stating that actions are monotone:

$$a(x + y) + a(x) = a(x + y).$$

The theory of simulation equivalence then yields a presentation of the graded monad with $M_n X$ defined inductively along with a partial ordering as follows: We take $M_0 X = \mathscr{P}_{\mathsf{f}}(X)$ ordered by set inclusion. We equip $\mathcal{A} \times M_n X$ with the product ordering of the discrete order on $\mathcal{A}$ and the given ordering on $M_n X$. Then $M_{n+1} X = \mathscr{P}^{\downarrow}_{\mathsf{f}}(\mathcal{A} \times M_n X)$ is the set of downwards-closed finite subsets of $\mathcal{A} \times M_n X$.

In the following lemma, an *epi-transformation* is a natural transformation α whose components α_X are surjective maps.

Lemma 2.11 [35]**.** *A graded monad $\mathbb{M}$ on* **Set** *is* depth-1 *if and only if all $\mu^{1,n}$ are epi-transformations and the following is object-wise a coequalizer diagram in the category of Eilenberg-Moore algebras for the monad M_0 for all $n \in \omega$:*

$$M_1 M_0 M_n \underset{\mu^{1,0} M_n}{\overset{M_1 \mu^{0,n}}{\rightrightarrows}} M_1 M_n \xrightarrow{\mu^{1,n}} M_{1+n}. \tag{2.7}$$

3. Graded Behavioural Equivalences

We next recall the notion of a graded semantics [35] on coalgebras for an endofunctor on **Set**; we illustrate several instantiations of subsequent interest.

Definition 3.1 (Graded semantics)**.** A *(depth-1) graded semantics* for an endofunctor $G\colon \mathbf{Set} \to \mathbf{Set}$ is a pair $(\alpha, \mathbb{M})$ consisting of a (depth-1) graded monad $\mathbb{M}$ on **Set** and a natural transformation $\alpha\colon G \to M_1$.

Given a G-coalgebra (X, γ), the graded semantics $(\alpha, \mathbb{M})$ induces a sequence of maps $\gamma^{(n)}\colon X \to M_n 1$ inductively defined by

$$\gamma^{(0)} := (X \xrightarrow{\eta X} M_0 X \xrightarrow{M_0 !} M_0 1);$$

$$\gamma^{(n+1)} := (X \xrightarrow{\alpha_X \cdot \gamma} M_1 X \xrightarrow{M_1 \gamma^{(n)}} M_1 M_n 1 \xrightarrow{\mu_1^{1,n}} M_{1+n} 1)$$

(or, using the graded Kleisli star, $\gamma^{(n+1)} = (\gamma^{(n)})_1^* \cdot \alpha_X \cdot \gamma$). We call $\gamma^{(n)}(x) \in M_n 1$ the *n-step $(\alpha, \mathbb{M})$-behaviour* of $x \in X$.

Definition 3.2 (Graded behavioural equivalence)**.** States $x \in X, y \in Y$ in G-coalgebras (X, γ) and (Y, δ) are *depth-n behaviourally equivalent* under $(\alpha, \mathbb{M})$ if $\gamma^{(n)}(x) = \delta^{(n)}(y)$, and *$(\alpha, \mathbb{M})$-behaviourally equivalent* if $\gamma^{(n)}(x) = \delta^{(n)}(y)$ for all $n \in \omega$. We refer to $(\alpha, \mathbb{M})$-behavioural equivalence as a *graded behavioural equivalence* or just a *graded equivalence*.

Example 3.3. We recall [15, Section 4] several graded equivalences, restricting primarily to LTS and PTS.

(1) For an endofunctor G on **Set**, finite-depth G-behavioural equivalence arises as the graded equivalence with $\mathbb{M} = \mathbb{M}_G$ and $\alpha = \mathsf{id}$, where $\mathbb{M}_G$ is the graded monad of Example 2.5.(1). By Remark 2.3, it follows that $(\mathsf{id}, \mathbb{M}_G)$ captures full coalgebraic bisimilarity in case G is finitary.

(2) Let (X, γ) be an LTS, let $x \in X$, and let $w \in \mathcal{A}^*$ be a finite word over $\mathcal{A}$. We write $x \xrightarrow{w} y$ if the state y can be reached on a path whose labels form the word w. A *finite trace* at $x \in X$ is a word $w \in \mathcal{A}^*$ such that $x \xrightarrow{w} y$ for some $y \in X$; the set of finite traces at x is denoted $\tau(x)$. States $x, y \in X$ are trace equivalent if $\tau(x) = \tau(y)$. Trace equivalence on finitely branching LTS is captured by the graded equivalence induced by $\mathbb{M} = \mathbb{M}_{\mathscr{P}_\mathsf{f}}(\mathcal{A})$ (cf. Example 2.5.(3)), again with $\alpha = \mathsf{id}$; replacing $\mathscr{P}_\mathsf{f}$ with $\mathscr{P}^+$ (or with $\mathscr{P}_\mathsf{f}^+$) yields trace equivalence on serial (and finitely branching) LTS.

(3) Probabilistic trace equivalence on PTS is the graded equivalence induced by $\mathbb{M} = \mathbb{M}_{\mathcal{D}_f}(\mathcal{A})$ (cf. Example 2.5.(3)) and $\alpha = \mathsf{id}$: The maps $\gamma^{(k)}$ equip states with distributions on length-k action words, and the induced equivalence identifies states x and y whenever these distributions coincide at x and y for all k.

(4) Simulation equivalence on LTS can also be construed as a graded equivalence by taking $\mathbb{M}$ to be the graded monad described in Example 2.10.(4), and

$$\begin{aligned} \alpha_X\colon \mathscr{P}_\mathsf{f}(\mathcal{A} \times X) &\to \mathscr{P}_f^{\downarrow}(\mathcal{A} \times \mathscr{P}_\mathsf{f} X) \\ S &\mapsto {\downarrow}\{(a, \{x\}) \mid (a, x) \in S\} \end{aligned}$$

where $\downarrow$ takes downsets.

Remark 3.4. It follows from the depth-1 presentations described in Example 2.10 that the graded semantics mentioned in Example 3.3 are depth-1.

4. Graded Algebras

Graded monads come equipped with graded analogues of both the Eilenberg-Moore and Kleisli constructions for ordinary monads. In particular, we have a notion of *graded algebra* [19, 35]:

Definition 4.1 (Graded algebra)**.** Let $k \in \omega$ and let $\mathbb{M}$ be a graded monad on a category $\mathscr{C}$. An *M_k-algebra* A consists of a family of $\mathscr{C}$-objects $(A_n)_{n \le k}$ (the *carriers*) and a family of $\mathscr{C}$-morphisms

$$(n + m \le k) \qquad a^{n,m}\colon M_n A_m \to A_{n+m}$$

(the *structure*) such that $a^{0,n} \cdot \eta_{A_n} = \mathsf{id}_{A_n}$ $(n \leq k)$ and

$$\begin{array}{ccc} M_n M_m A_r & \xrightarrow{M_n a^{m,r}} & M_n A_{m+r} \\ {\scriptstyle \mu^{n,m}}\downarrow & & \downarrow{\scriptstyle a^{n,m+r}} \\ M_{n+m} A_r & \xrightarrow{a^{n+m,r}} & A_{n+m+r} \end{array} \tag{4.1}$$

for all $n, m, r \in \omega$ such that $n+m+r \leq k$. The *i-part* of an M_k-algebra A is the M_0-algebra $(A_i, a^{0,i})$.

A *homomorphism* from A to an M_k-algebra B is a family of $\mathscr{C}$-morphisms $h_n \colon A_n \to B_n$ $(n \leq k)$ such that

$$h_{n+m} \cdot a^{n,m} = b^{n,m} \cdot M_n h_m \quad \text{for all } n, m \in \omega \text{ s.th. } n+m \leq k.$$

We write $\mathsf{Alg}_k(\mathbb{M})$ for the category of M_k-algebras and their homomorphisms.

We define *M_ω-algebras* (and their homomorphisms) similarly, by allowing the indices n, m, r to range over ω.

Remark 4.2. The above notion of M_ω-algebra corresponds with the concept of graded Eilenberg-Moore algebras introduced by Fujii et al. [19]. Intuitively, M_ω-algebras are devices for interpreting terms of unbounded uniform depth. We understand M_k-algebras [35] as a refinement of M_ω-algebras which allows the interpretation of terms of uniform depth *at most* k. Thus, M_k-algebras serve as a formalism for specifying properties of states exhibited in k *steps*. For example, M_1-algebras are used to interpret one-step modalities of characteristic logics for graded semantics [15, 18]. Moreover, for a depth-1 graded monad, its M_ω-algebras may be understood as compatible chains of M_1-algebras [35], and a depth-1 graded monad can be reconstructed from its M_1-algebras.

We will be chiefly interested in M_0- and M_1-algebras:

Example 4.3. Let $\mathbb{M}$ be a graded monad on **Set**.

(1) An M_0-algebra is just an Eilenberg-Moore algebra for the monad $(M_0, \eta, \mu^{0,0})$. It follows that $\mathsf{Alg}_0(\mathbb{M})$ is complete and cocomplete, in particular has coequalizers.

(2) An M_1-algebra is a pair $((A_0, a^{0,0}), (A_1, a^{0,1}))$ of M_0-algebras – often we just write the carriers A_i to also denote the algebras, by abuse of notation – equipped with a *main structure map* $a^{1,0} \colon M_1 A_0 \to A_1$ satisfying two instances of (4.1). One instance states that $a^{1,0}$ is an M_0-algebra homomorphism from $(M_1 A_0, \mu_A^{0,1})$ to $(A_1, a^{0,1})$ (*homomorphy*); the other expresses that $a^{1,0} \cdot \mu^{1,0} = a^{1,0} \cdot M_1 a^{0,0}$ (*coequalization*):

$$M_1 M_0 A_0 \underset{M_1 a^{0,0}}{\overset{\mu^{1,0}}{\rightrightarrows}} M_1 A_0 \xrightarrow{a^{1,0}} A_1. \tag{4.2}$$

Remark 4.4. The free M_n-algebra on a set X is formed in the expected way, in particular has carriers $M_0 X, \ldots, M_n X$, see [35, Prop. 6.3].

Canonical algebras. We are going to review the basic definitions and results on *canonical M_1-algebras* [15]. Fix a graded monad $\mathbb{M}$ on **Set**.

We write $(-)_i \colon \mathsf{Alg}_1(\mathbb{M}) \to \mathsf{Alg}_0(\mathbb{M})$, $i = 0, 1$, for the functor which sends an M_1-algebra A to its i-part A_i and sends a homomorphism $h \colon A \to B$ to $h_i \colon A_i \to B_i$.

Definition 4.5. An M_1-algebra A is *canonical* if it is free over its 0-part with respect to $(-)_0 \colon \mathsf{Alg}_1(\mathbb{M}) \to \mathsf{Alg}_0(\mathbb{M})$.

Remark 4.6. The universal property of a canonical algebra A is the following: for every M_1-algebra B and every M_0-algebra homomorphism $h \colon A_0 \to B_0$, there exists a unique M_1-algebra homomorphism $h^\# \colon A \to B$ such that $(h^\#)_0 = h_0$.

Lemma 4.7 [15, Lem. 5.3]**.** *An M_1-algebra A is canonical if and only if (4.2) is a coequalizer in $\mathsf{Alg}_0(\mathbb{M})$.*

Example 4.8. Let X be a set and let $\mathbb{M}$ be a depth-1 graded monad on **Set**. For each $k \in \omega$, we may view $M_k X$ as an M_0-algebra with structure $\mu^{0,k}$. For the M_1-algebra $(M_k X, M_{k+1} X)$ (with main structure map $\mu^{1,k}$), the instance of Diagram (4.2) required by Lemma 4.7 is a coequalizer by Lemma 2.11; that is, $(M_k X, M_{k+1} X, \mu^{1,k})$ is canonical.

5. Pre-Determinization in Eilenberg-Moore

We describe a generic notion of pre-determinization (the terminology will be explained in Remark 5.2) for coalgebras of an endofunctor G on **Set** with respect to a given depth-1 graded semantics $(\alpha, \mathbb{M})$, generalizing the Eilenberg-Moore-style coalgebraic determinization construction by Silva et al. [38]. The behavioural equivalence game introduced in the next section will effectively be played on the pre-determinization of the given coalgebra. We will occasionally gloss over issues of finite branching in the examples.

We first note that every M_0-algebra A extends (uniquely) to a canonical M_1-algebra EA (with 0-part A), whose 1-part and main structure are obtained by taking the coequalizer of the pair of morphisms in (4.2) (canonicity then follows by Lemma 4.7). This construction forms the object part of a functor $\mathsf{Alg}_0(\mathbb{M}) \to \mathsf{Alg}_1(\mathbb{M})$ which sends a homomorphism $h\colon A \to B$ to its unique extension $Eh := h^\sharp\colon EA \to EB$ (cf. Remark 4.6). We write $\overline{M}_1$ for the endofunctor on $\mathsf{Alg}_0(\mathbb{M})$ given by

$$\overline{M}_1 := (\mathsf{Alg}_0(\mathbb{M}) \xrightarrow{E} \mathsf{Alg}_1(\mathbb{M}) \xrightarrow{(-)_1} \mathsf{Alg}_0(\mathbb{M})), \tag{5.1}$$

where $(-)_1$ is the functor taking 1-parts. Thus, for an M_0-algebra A_0, $\overline{M}_1(A_0)$ is the vertex of the coequalizer (4.2).

By Example 4.8, we have

$$\overline{M}_1(M_kX, \mu_X^{0,k}) = (M_{k+1}X, \mu_X^{0,k+1}) \tag{5.2}$$

for every set X and every $k \in \omega$. In particular,

$$U\overline{M}_1F = M_1 \tag{5.3}$$

where $F \dashv U\colon \mathsf{Alg}_0(\mathbb{M}) \to \mathbf{Set}$ is the canonical adjunction of the Eilenberg-Moore category of M_0 – that is, U is the forgetful functor, and F takes free M_0-algebras, so $FX = (M_0X, \mu_X^{00})$. For an M_1-coalgebra $f\colon X \to M_1X = U\overline{M}_1FX$, we therefore obtain a homomorphism $f^\#\colon FX \to \overline{M}_1FX$ (in $\mathsf{Alg}_0(\mathbb{M})$) via adjoint transposition. This leads to the following pre-determinization construction:

Definition 5.1. Let $(\alpha, \mathbb{M})$ be a depth-1 graded semantics on G-coalgebras. The *pre-determinization* of a G-coalgebra (X, γ) under $(\alpha, \mathbb{M})$ is the $\overline{M}_1$-coalgebra

$$(\alpha_X \cdot \gamma)^\#\colon FX \to \overline{M}_1FX. \tag{5.4}$$

Remark 5.2. (1) We call this construction a *pre*-determinization because it will serve as a *determinization* – in the expected sense that the underlying graded equivalence transforms into behavioural equivalence on the determinization – only under additional conditions. Notice that given a G-coalgebra (X, γ), (finite-depth) behavioural equivalence on the $\overline{M}_1$-coalgebra $(\alpha_X \cdot \gamma)_0^*$ is given by the canonical cone into the final chain

$$1 \xleftarrow{!} \overline{M}_1 1 \xleftarrow{\overline{M}_1!} \overline{M}_1^2 1 \xleftarrow{\overline{M}_1^2!} \cdots$$

while graded behavioural equivalence on (X, γ) is given by the maps $\gamma^{(k)}$ into the sequence $M_01, M_11, M_21, \ldots$, equivalently given as homomorphisms $(\gamma^{(k)})_0^*\colon FX \to (M_k1, \mu_1^{0,k})$, whose codomains can, by (5.2), be written as the sequence

$$F1, \quad \overline{M}_11, \quad \overline{M}_1^21, \quad \ldots$$

of M_0-algebras. The two sequences coincide in case $M_01 = 1$, and indeed one easily verifies that in this case, finite-depth behavioural equivalence on $\overline{M}_1$-coalgebras coincides with $(\alpha, \mathbb{M})$-behavioural equivalence. For instance, this holds in the case of probabilistic trace equivalence (Example 3.3.(3)), where $M_0 = \mathcal{D}$, so $M_01 = 1$. In the case of trace equivalence (Example 3.3.(2)), $M_01 = 1$ can be ensured by restricting to serial labelled transition systems, which, as noted in Example 2.2.(1), are coalgebras for $\mathscr{P}^+(\mathcal{A}\times -)$ with $\mathscr{P}^+$ denoting non-empty powerset, so that in the corresponding variant of the graded monad for trace semantics, we have $M_0 = \mathscr{P}^+$ and hence $M_01 = 1$.

On the other hand, the condition $M_01 = 1$ fails for trace equivalence of unrestricted systems where we have $M_0 = \mathscr{P}$, which in fact constitutes a radical example where behavioural equivalence on the pre-determization is strictly coarser than the given graded equivalence. In this case, since the actions preserve the bottom 0, we in fact have $\overline{M}_11 = 1$: it follows that *all* states in $\overline{M}_1$-coalgebras are behaviourally equivalent (as the unique coalgebra structure on 1 is final).

(2) Using (5.3), we see that the underlying map of the pre-determinization of a coalgebra (X,γ) is $(\alpha_X\cdot\gamma)_0^*\colon M_0X \to M_1X = U_0\overline{M}_1F_0X$ (written using graded Kleisli star as per Notation 2.6). Indeed, one easily shows that $(\alpha_X\cdot\gamma)_0^*$ is an M_0-algebra morphism $(M_0X,\mu_X^{0,0}) \to \overline{M}_1(M_0X,\mu_X^{0,0}) = (M_1X,\mu_X^{0,1})$ satisfying $(\alpha_X\cdot\gamma)_0^*\cdot\eta_X = \alpha_X\cdot\gamma$. Thus, it is the adjoint transpose in (5.4).

(3) As indicated above, pre-determinization captures the Eilenberg-Moore style generalized determinization by Silva et al. [38] as an instance. Indeed, for a monad T and an endofunctor F, both on the category $\mathscr{C}$, one considers a coalgebra $\gamma\colon X \to FTX$. Assuming that FTX carries the structure of an Eilenberg-Moore algebra for T (e.g. because the functor F lifts to the category of Eilenberg-Moore algebras for T), one obtains an F-coalgebra $\gamma^\sharp\colon TX \to FTX$ by taking the unique homomorphic extension of γ. Among the concrete instances of this construction are the well-known powerset construction of non-deterministic automata (take $T=\mathscr{P}$ and $F = 2\times(-)^A$), the non-determinization of alternating automata and that of Markov decision processes [26].

To view this as an instance of pre-determinization, take the graded monad with $M_n = F^nT$ (Example 2.5.(4)), let $G = FT$, and let $\alpha = \mathsf{id}_{FT}$. Using (5.3), we see that $(\alpha_X\cdot\gamma)^\#$ in (5.4) is the generalized determinization $\gamma^\sharp$ above.

(4) We emphasize that the construction applies completely universally; e.g. we obtain as one instance a 'determinization' of serial labelled transition systems modulo similarity, which transforms a coalgebra $X \to \mathscr{P}^+(\mathcal{A}\times X)$ into an $\overline{M}_1$-coalgebra $\mathscr{P}^+(X) \to \mathscr{P}^\downarrow(\mathcal{A}\times\mathscr{P}^+(X))$ (Example 2.10.(4)); instantiating the observations in item (1), we obtain that finite-depth behavioural equivalence of $\overline{M}_1$-coalgebras (see Example 5.3 for the description of $\overline{M}_1$) coincides with finite-depth mutual similarity.

Example 5.3. We give a description of the functor $\overline{M}_1$ on M_0-algebras constructed above in some of the running examples.

(1) For graded monads of the form $\mathbb{M}_G$, which capture finite-depth behavioural equivalence (Example 2.5.(1)), we have $M_0 = \mathsf{Id}$, so M_0-algebras are just sets, and under this correspondence, $\overline{M}_1$ is the original functor G.

(2) Trace semantics of LTS (Example 2.10.(3)): Distribution of actions over the join semilattice operations ensures that depth-1 terms over a join semilattice X can be normalized to sums of the form $\sum_{a\in\mathcal{A}} a(x_a)$, with $x_a \in X$ (possibly $x_a = 0$). It follows that $\overline{M}_1$ is simply given by $\overline{M}_1X = X^{\mathcal{A}}$ ($\mathcal{A}$-th power, where $\mathcal{A}$ is the finite set of labels). Other forms of trace semantics are treated similarly.

(3) In the graded theory for simulation (Example 2.10.(4)), the description of the induced graded monad [15] extends analogously to $\overline{M}_1$, yielding that $\overline{M}_1B$ is the join semilattice of finitely generated downwards closed subsets of $\mathcal{A}\times B$ where, again, $\mathcal{A}$ carries the discrete ordering.

Remark 5.4. The assignment $\mathbb{M} \mapsto \overline{M}_1$ exhibits the category $\mathscr{K}$ of depth-1 graded monads whose 0-part is the monad $(M_0,\eta,\mu^{0,0})$ as a coreflective subcategory (up to isomorphism) of the category $\mathsf{Fun}(\mathbf{Set}^{M_0})$ of all endofunctors on the Eilenberg-Moore category of that monad.

Indeed, given an endofunctor H on $\mathbf{Set}^{M_0}$ we form the 6-tuple $(M_0, UHF, \eta, \mu^{0,0}, \mu^{0,1}, \mu^{1,0})$, where the latter two natural transformations arise from the counit $\varepsilon\colon FU \to \mathsf{Id}$ of the canonical adjunction $F\dashv U\colon \mathsf{Alg}_0(\mathbb{M}_0) \to \mathbf{Set}$:

$$\mu^{0,1} = \big(M_0UHF = UFUHF \xrightarrow{U\varepsilon HF} UHF\big);$$
$$\mu^{1,0} = \big(UHFM_0 = UHFUF \xrightarrow{UHF\varepsilon F} UHF\big).$$

It is not difficult to check that this data satisfies all applicable instances of the graded monad laws. Hence, it specifies a depth-1 graded monad $R(H)$ [15, Thm. 3.7]; this assignment is the object part of a functor $R\colon \mathsf{Fun}(\mathbf{Set}^{M_0}) \to \mathscr{K}$.

In the other direction, we have for each depth-1 graded monad $\mathbb{M}$ with 0-part M_0 the endofunctor $I(\mathbb{M}) = \overline{M}_1$. By (5.3), we have $RI(\mathbb{M}) = \mathbb{M}$. Now, given a depth-1 graded monad $\mathbb{M}$ and an endofunctor H on $\mathbf{Set}^{M_0}$, consider $\overline{M}_1 = IR(H)$ (so that $M_1 = UHF$). We obtain for every algebra (A,a) in $\mathbf{Set}^{M_0}$ a homomorphism $c_{(A,a)}\colon \overline{M}_1(A,a) \to H(A,a)$ by using the coequalizer defining $\overline{M}_1(A,a)$

(cf. Lemma 4.7):

$$\begin{array}{ccccc} M_1M_0A & \overset{\mu_A^{0,1}}{\underset{M_1a}{\rightrightarrows}} & M_1A = HFA & \twoheadrightarrow & \overline{M}_1(A,a) \\ & & & \searrow^{Ha} & \downarrow{\scriptstyle c_{(A,a)}} \\ & & & & H(A,a) \end{array}$$

Note that M_1M_0A is the carrier of the Eilenberg-Moore algebra $HFA = H(M_0A, \mu_A^{0,0})$ and similarly for the middle object (in both cases we have omitted the algebra structures given by $\mu_{M_0A}^{0,1}$ and $\mu_A^{0,1}$ coming from the graded monad $I(H)$). It is easy to see that the homomorphism Ha merges the parallel pair, and therefore we obtain the dashed morphism such that the triangle commutes, yielding the components of a natural transformation $c\colon \overline{M}_1 \to H$ which is couniversal: for each depth-1 graded monad $\mathbb{N}$ whose 0-part is M_0 and each natural transformation $h\colon \overline{M}_1 \to H$, there is a unique natural transformation $m_1\colon N_1 \to M_1 = UHF$ such that $m = (id_{M_0}, m_1)$ is a morphism of graded monads from $\mathbb{N}$ to $\mathbb{M}$ and $c \cdot I(m) = h$. This shows that $I \dashv R$.

6. Behavioural Equivalence Games

Let $\mathcal{S} = (\alpha, \mathbb{M})$ be a depth-1 graded semantics for an endofunctor G on **Set**. We are going to describe a game for playing out depth-n behavioural equivalence under $\mathcal{S}$-semantics on states in G-coalgebras.

We first give a description of the game in the syntactic language of graded equational reasoning, and then present a more abstract categorical definition. Given a coalgebra (X, γ), we will see the states in X as variables, and the map $\alpha_X \cdot \gamma$ as assigning to each variable x a depth-1 term over X; we can regard this assignment as a (uniform-depth) substitution σ. A configuration of the game is a pair of depth-0 terms over X; to play out the equivalence of states $x, y \in X$, the game is started from the initial configuration (x, y). Each round of the game then proceeds in two steps: First, Duplicator plays a set Z of equalities between depth-0 terms over X that she claims to hold under the semantics. This move is admissible in the configuration (s, t) if $Z \vdash s\sigma = t\sigma$. Then, Spoiler challenges one of the equalities claimed by Duplicator, i.e. picks an element $(s', t') \in Z$, which then becomes the next configuration. Any player who cannot move, loses. After n rounds have been played, reaching the final configuration (s, t), Duplicator wins if $s\theta = t\theta$ is a valid equality, where θ is a substitution that identifies all variables. We refer to this last check as *calling the bluff*. Thus, the game plays out an equational proof between terms obtained by unfolding depth-0 terms according to σ, cutting off after n steps.

We introduce some technical notation to capture the notion of admissibility of Z abstractly:

Notation 6.1. Let $Z \subseteq M_0X \times M_0X$ be a relation, and let $c_Z\colon M_0X \to C_Z$ be the coequalizer in $\mathsf{Alg}_0(\mathbb{M})$ of the homomorphisms $\ell_0^*, r_0^*\colon M_0Z \to M_0X$ given by applying the Kleisli star (2.3) to the projections $\ell, r\colon Z \to M_0X$. We define a homomorphism $\mathcal{Z}\colon M_0X \to M_1C_Z$ in $\mathsf{Alg}_0(\mathbb{M})$ by

$$\mathcal{Z} = \big(M_0X \xrightarrow{(\alpha_X \cdot \gamma)_0^*} M_1X = \overline{M}_1M_0X \xrightarrow{\overline{M}_1c_Z} \overline{M}_1C_Z\big) \tag{6.1}$$

(omitting algebra structures, and again using the Kleisli star).

Remark 6.2. Using designators as in Notation 6.1, we note:

(1) By the universal property of $\eta_Z\colon Z \to M_0Z$, an M_0-algebra homomorphism $h\colon M_0X \to A$ merges ℓ, r iff it merges ℓ_0^*, r_0^*. This implies that the coequalizer $M_0X \xrightarrow{c_Z} C_Z$ quotients the free M_0-algebra M_0X by the congruence generated by Z. Also, it follows that in case Z is already an M_0-algebra and $\ell, r\colon Z \to M_0X$ are M_0-algebra homomorphisms (e.g. when Z is a congruence), one may take $c_Z\colon M_0X \to C_Z$ to be the coequalizer of ℓ, r.

(2) The map $\mathcal{Z}\colon M_0X \to \overline{M}_1C_Z$ associated to the relation Z on M_0X may be understood as follows. As per the discussion above, we view the states of the coalgebra (X, γ) as variables, and the map $X \xrightarrow{\gamma} GX \xrightarrow{\alpha_X} M_1X$ as a substitution mapping a state $x \in X$ to the equivalence class of depth-1 terms encoding the successor structure $\gamma(x)$. The second factor $\overline{M}_1c_Z$ in (6.1) then essentially applies the relations given by the closure of Z under congruence w.r.t. depth-0 operations, embodied in c_Z as per (1), under depth-1 operations in (equivalence classes of) of depth-1 terms in M_1X; to sum up, $\overline{M}_1c_Z$ merges a pair of equivalence classes $[t], [t']$ iff $Z \vdash t = t'$ in a depth-1 theory presenting $\mathbb{M}$ (in notation as per Section 2).

Definition 6.3. For $n \in \omega$, the *n-round $\mathcal{S}$-behavioural equivalence game* $\mathcal{G}_n(\gamma)$ on a G-coalgebra (X, γ) is played by Duplicator (D) and Spoiler (S). *Configurations* of the game are pairs $(s, t) \in M_0(X) \times M_0(X)$. Starting from an *initial configuration* designated as needed, the game is played for n rounds. Each round proceeds in two steps, from the current configuration (s, t): First, D chooses a relation $Z \subseteq M_0X \times M_0X$ such that $Z(s) = Z(t)$ (for Z as per Notation 6.1). Then, S picks an element $(s', t') \in Z$, which becomes the next configuration. Any player who cannot move at his turn, loses. After n rounds have been played, D wins if $M_0!(s_n) = M_0!(t_n)$; otherwise, S wins.

Remark 6.4. By the description of Z given in Remark 6.2.(2), the categorical definition of the game corresponds to the algebraic one given in the lead-in discussion. The final check whether $M_0!(s_n) = M_0!(t_n)$ corresponds to what we termed *calling the bluff*. The apparent difference between playing either on depth-0 terms or on elements of M_0X, i.e. depth-0 terms modulo derivable equality, is absorbed by equational reasoning from Z, which may incorporate also the application of depth-0 equations.

Remark 6.5. A pair of states coming from different coalgebras (X, γ) and (Y, δ) can be treated by considering those states as elements of the coproduct of the two coalgebras:

$$X + Y \xrightarrow{\gamma+\delta} GX + GY \xrightarrow{[G\mathsf{inl}, G\mathsf{inr}]} G(X + Y),$$

where $X \xrightarrow{\mathsf{inl}} X + Y \xleftarrow{\mathsf{inr}} Y$ denote the coproduct injections. There is an evident variant of the game played on two different coalgebras (X, γ), (Y, δ), where moves of D are subsets of $M_0X \times M_0Y$. However, completeness of this version depends on additional assumptions on $\mathbb{M}$, to be clarified in future work. For instance, if we instantiate the graded monad for traces with effects specified by T (Example 2.5.(3)) to T being the free real vector space monad, and a state $x \in X$ has successor structure $2 \cdot x' - 2 \cdot x''$, then D can support equivalence between x and a deadlock $y \in Y$ (with successor structure 0) by claiming that $x' = x''$, but not by any equality between terms over X with terms over Y. That is, in this instance, the variant of the game where D plays relations on $M_0X \times M_0Y$ is not complete.

Soundness and completeness of the game with respect to $\mathcal{S}$-behavioural equivalence is stated as follows.

Theorem 6.6. *Let $(\alpha, \mathbb{M})$ be a depth-1 graded semantics for a functor G such that $\bar{M}_1$ preserves monomorphisms, and let (X, γ) be a G-coalgebra. Then, for all $n \in \omega$, D wins (s, t) in $\mathcal{G}_n(\gamma)$ if and only if $(\gamma^{(n)})^*_0(s) = (\gamma^{(n)})^*_0(t)$.*

Corollary 6.7. *States x, y in a G-coalgebra (X, γ) are $\mathcal{S}$-behaviourally equivalent if and only if D wins $(\eta(x), \eta(y))$ for all $n \in \omega$.*

Remark 6.8. In algebraic terms, the condition that $\bar{M}_1$ preserves monomorphisms amounts to the following: In the derivation of an equality of depth-1 terms s, t over X from depth-0 relations over X (i.e. from a presentation of an M_0-algebra by relations on generators X), if X is included in a larger set Y of variables with relations that conservatively extend those on X, i.e. do not imply additional relations on X, then it does not matter whether the derivation is conducted over X or more liberally over Y. Intuitively, this property is needed because not all possible n-step behaviours, i.e. elements of $Y = M_n1$, are realized by states in a given coalgebra on X. Preservation of monos by $\bar{M}_1$ is automatic for graded monads of the form $\mathbb{M}_G$ (Example 2.5.(1)), since $M_0 = \mathsf{Id}$ in this case. In the other running examples, preservation of monos is by the respective descriptions of $\bar{M}_1$ given in Example 5.3.

Example 6.9. We take a brief look at the instance of the generic game for the case of bisimilarity on finitely branching LTS (more extensive examples are in Section 8), i.e. we consider the depth-1 graded semantics $(\mathsf{id}, \mathbb{M}_G)$ for the functor $G = \mathscr{P}_\mathsf{f}(\mathcal{A} \times (-))$. In this case, $M_0 = \mathsf{Id}$, so when playing on a coalgebra (X, γ), D plays relations $Z \subseteq X \times X$. If the successor structures of states x, y are represented by depth-1 terms $\sum_i a_i(x_i)$ and $\sum_j b_j(y_j)$, respectively, in the theory $\mathsf{JSL}(\mathcal{A})$ (Example 2.10.(1)), then D is allowed to play Z iff the equality $\sum_i a_i(x_i) = \sum_j b_j(y_j)$ is entailed by Z in $\mathsf{JSL}(\mathcal{A})$. This, in turn, holds iff for each i, there is j such that $a_i = b_j$ and $(x_i, y_j) \in Z$, and symmetrically. Thus Z may be seen as a pre-announced non-deterministic winning strategy for D in the usual bisimilarity game where S moves first (Section 2): D announces that if S moves from, say, x to x_i, then she will respond with some y_j such that $a_i = b_j$ and $(x_i, y_j) \in Z$.

7. Infinite-depth behavioural equivalence

We have seen in Section 5 that in case $M_0 1 = 1$, $(\alpha, \mathbb{M})$-behavioural equivalence on G-coalgebras coincides, via a determization construction, with finite-depth behavioural equivalence on $\bar{M}_1$-coalgebras for a functor $\bar{M}_1$ on M_0-algebras constructed from $\mathbb{M}$. If G is finitary, then finite-depth behavioural equivalence coincides with full behavioural equivalence (Remark 2.3), but in general, finite-depth behavioural equivalence is strictly coarser. Previous treatments of graded semantics stopped at this point, in the sense that for non-finitary functors (which describe infinitely branching systems), they did not offer a handle on infinite-depth equivalences such as full bisimilarity. In case $M_0 1 = 1$, a candidate for a notion of infinite-depth equivalence induced by a graded semantics arises via full behavioural equivalence of $\bar{M}_1$-coalgebras. We fix this notion explicitly:

Definition 7.1. States x, y in a G-coalgebra (X, γ) are *infinite-depth $(\alpha, \mathbb{M})$-behaviourally equivalent* if $\eta(x)$ and $\eta(y)$ are behaviourally equivalent in the pre-determinization of (X, γ) as described in Section 6.

We hasten to re-emphasize that this notion in general only makes sense in case $M_0 1 = 1$. We proceed to show that infinite-depth equivalence is in fact captured by an infinite variant of the behavioural equivalence game of Section 6.

Since infinite depth-equivalences differ from finite-depth ones only in settings with infinite branching, we do not assume in this section that G or $\mathbb{M}$ are finitary, and correspondingly work with generalized graded theories where operations may have infinite arities [35]; we assume arities to be cardinal numbers. We continue to be interested only in depth-1 graded monads and theories, and we fix such a graded monad $\mathbb{M}$ and associated graded theory for the rest of this section. The notion of derivation is essentially the same as in the finitary case, the most notable difference being that the congruence rule is now infinitary, as it has one premise for each argument position of a given possibly infinitary operator. We do not impose any cardinal bound on the arity of operations; if all operations have arity less than κ for a regular cardinal κ, then we say that the monad is *κ-ary*.

Remark 7.2. One can show using tools from the theory of locally presentable categories that $\bar{M}_1$ has a final coalgebra if $\mathbb{M}$ is κ-ary in the above sense. To see this, first note that $\mathsf{Alg}_0(\mathbb{M})$ is locally κ-presentable if M_0 is κ-accessible [6, Remark 2.78]. Using a somewhat similar argument one can prove that $\mathsf{Alg}_1(\mathbb{M})$ is also locally κ-presentable. Moreover, the functor $\bar{M}_1$ is κ-accessible, being the composite (5.1) of the left adjoint $E\colon \mathsf{Alg}_0(\mathbb{M}) \to \mathsf{Alg}_1(\mathbb{M})$ (which preserves all colimits) and the 1-part functor $(-)_1\colon \mathsf{Alg}_1(\mathbb{M}) \to \mathsf{Alg}_0(\mathbb{M})$, which preserves κ-filtered colimits since those are formed componentwise. It follows that $\bar{M}_1$ has a final coalgebra [6, Exercise 2j]. Alternatively, existence of a final $\bar{M}_1$-coalgebra will follow from Theorem 7.12 below.

Like before, we *assume that $\bar{M}_1$ preserves monomorphisms.*

Example 7.3. We continue to use largely the same example theories as in Example 2.10, except that we allow operations to be infinitary. For instance, the *graded theory of complete join semilattices over $\mathcal{A}$* has as depth-1 operations all formal sums $\sum_{i\in I} a_i(-)$ where I is now some (possibly infinite) index set; the axioms are then given in the same way as in Example 2.10.(1), and all depth-1 equations

$$\textstyle\sum_{i\in I} a_i(x) = \sum_{j\in J} b_j(y)$$

such that $\{(a_i, x_i) \mid i \in I\} = \{(b_j, y_j) \mid j \in J\}$. This theory presents the graded monad $\mathbb{M}_G$ for $G = \mathscr{P}(\mathcal{A} \times (-))$.

The infinite game may then be seen as defining a notion of derivable equality on infinite-depth terms by playing out a non-standard, infinite-depth equational proof; we will make this view explicit further below. In a less explicitly syntactic version, the game is defined as follows.

Definition 7.4 (Infinite behavioural equivalence game)**.** The *infinite $(\alpha, \mathbb{M})$-behavioural equivalence game* $\mathcal{G}_\infty(\gamma)$ on a G-coalgebra (X, γ) is played by Spoiler (S) and Duplicator (D) in the same way as the finite behavioural equivalence game (Definition 6.3) except that the game continues forever unless one of the players cannot move. Any player who cannot move, loses. Infinite matches are won by D.

As indicated above, this game captures infinite-depth $(\alpha, \mathbb{M})$-behavioural equivalence (under the running assumption that $\bar{M}_1$ preserves monomorphisms):

Theorem 7.5. *Given a G-coalgebra (X, γ), two states s, t in the pre-determinization of γ are behaviourally equivalent iff D wins the infinite $(\alpha, \mathbb{M})$-behavioural equivalence game $\mathcal{G}_\infty(\gamma)$ from the initial configuration (s, t).*

Corollary 7.6. *Two states x, y in a G-coalgebra (X, γ) are infinite-depth $(\alpha, \mathbb{M})$-behaviourally equivalent iff D wins the infinite $(\alpha, \mathbb{M})$-behavioural equivalence game $\mathcal{G}_\infty(\gamma)$ from the initial configuration $(\eta(x), \eta(y))$.*

Remark 7.7. Like infinite-depth $(\alpha, \mathbb{M})$-behavioural equivalence, the infinite $(\alpha, \mathbb{M})$-behavioural equivalence game is sensible only in case $M_0 1 = 1$. For instance, as noted in Section 5, in the graded monad for trace semantics (Example 2.10.(2)), which does not satisfy this condition, behavioural equivalence of $\bar{M}_1$-coalgebras is trivial. In terms of the game, D wins every position in $\mathcal{G}_\infty(\gamma)$ by playing $Z = \{(t, 0) \mid t \in M_0 X\}$ – since the actions preserve the bottom element 0, this is always an admissible move. In the terminology introduced at the beginning of Section 6, the reason that D wins in this way is that in the infinite game, her bluff is never called ($M_0!(t)$ will in general not equal $M_0!(0) = 0$). However, see Example 7.8.(1) below.

Example 7.8. (1) As noted in Remark 5.2.(1), the graded monad for trace semantics can be modified to satisfy the condition $M_0 1 = 1$ by restricting to serial labelled transition systems. By modification of Example 5.3.(2), we obtain that in this setting, $\bar{M}_1 X$ consists of partial maps $\mathcal{A} \rightharpoonup X$ that are not everywhere undefined. It follows that in this case, infinite-depth $(\alpha, \mathbb{M})$-behavioural equivalence is just finite trace equivalence, and thus coincides with plain $(\alpha, \mathbb{M})$-behavioural equivalence.

(2) In the case of graded monads $\mathbb{M}_G$ (Example 2.5.(1)), which so far were used to capture finite-depth behavioural equivalence in the standard (branching-time) sense, we have $M_0 = \mathsf{Id}$; in particular, $M_0 1 = 1$. In this case, the infinite-depth behavioural equivalence game instantiates to a game that characterizes full behavioural equivalence of G-coalgebras. Effectively, a winning strategy of D in the infinite game $\mathcal{G}_\infty(\gamma)$ on a G-coalgebra (X, γ) amounts to a relation $R \subseteq X \times X$ (the positions of D actually reachable when D follows her winning strategy) that is a *precongruence* on (X, γ) [3].

Remark 7.9 (Fixpoint computation)**.** Via its game characterization (Theorem 7.5), infinite-depth $(\alpha, \mathbb{M})$-behavioural equivalence can be cast as a greatest fixpoint, specifically of the monotone function F on $\mathscr{P}(M_0 X \times M_0 X)$ given by

$$F(Z) = \{(s, t) \in M_0 X \times M_0 X \mid Z(s) = Z(t)\}.$$

If M_0 preserves finite sets, then this fixpoint can be computed on a finite coalgebra (X, γ) by fixpoint iteration; since $F(Z)$ is clearly always an equivalence relation, the iteration converges after at most $|M_0 X|$ steps, e.g. in exponentially many steps in case $M_0 = \mathscr{P}$. In case $M_0 X$ is infinite (e.g. if $M_0 = \mathcal{D}$), then one will need to work with finite representations of subspaces of $M_0 X \times M_0 X$. We leave a more careful analysis of the algorithmics and complexity of solving infinite $(\alpha, \mathbb{M})$-behavioural equivalence games to future work. We do note that on finite coalgebras, we may assume w.l.o.g. that both the coalgebra functor G and graded monad $\mathbb{M}$ are finitary, as we can replace them with their finitary parts if needed (e.g. the powerset functor $\mathscr{P}$ and the finite powerset functor $\mathscr{P}_\mathsf{f}$ have essentially the same finite coalgebras). If additionally $M_0 1 = 1$, then $(\alpha, \mathbb{M})$-behavioural equivalence coincides with infinite-depth $(\alpha, \mathbb{M})$-behavioural equivalence, so that we obtain also an algorithmic treatment of $(\alpha, \mathbb{M})$-behavioural equivalence. By comparison, such a treatment is not immediate from the finite version of the game, in which the number of rounds is effectively chosen by Spoiler in the beginning.

Assume from now on that $\mathbb{M}$ is κ-ary. We note that in this case, we can describe the final $\bar{M}_1$-coalgebra in terms of a syntactic variant of the infinite game that is played on infinite-depth terms, defined as follows.

Definition 7.10 (Infinite-depth terms)**.** Recall that we are assuming a graded signature Σ with operations of arity less than κ. A *(uniform) infinite-depth (Σ-)term* is an infinite tree with ordered branching where each node is labelled with an operation $f \in \Sigma$, and then has as many children as given by the arity of f; when there is no danger of confusion, we will conflate nodes with (occurrences of) operations. We require moreover that every infinite path in the tree contains infinitely many depth-1 operations (finite full paths necessarily end in constants). We write $\mathscr{T}_{\Sigma,\infty}$ for the set of infinite-depth Σ-terms. By cutting off at the top-most depth-1 operations, we obtain for every $t \in \mathscr{T}_{\Sigma,\infty}$ a *top-level decomposition* $t = t_1 \sigma$ into a depth-1 term $t_1 \in \mathscr{T}_{\Sigma,1}(X)$, for some set X, and a substitution $\sigma \colon X \to \mathscr{T}_{\Sigma,\infty}$.

Definition 7.11. The *syntactic infinite* $(\alpha, \mathbb{M})$*-behavioural equivalence game* $\mathcal{G}_\infty^{\mathsf{syn}}$ is played by S and D. Configurations of the game are pairs (s,t) of infinite-depth Σ-terms. For such (s,t), we can assume, by the running assumption that $\bar{M}_1$ preserves monomorphisms, that the top level decompositions $s = s_1\sigma$, $t = t_1\sigma$ are such that $s_1, t_1 \in \mathscr{T}_{\Sigma,1}(X)$, $\sigma\colon X \to \mathscr{T}_{\Sigma,\infty}$ for the same X, σ. Starting from a designated initial configuration, the game proceeds in rounds. In each round, starting from a current such configuration (s,t), D first chooses a relation $Z \subseteq \mathscr{T}_{\Sigma,0}(X) \times \mathscr{T}_{\Sigma,0}(X)$ such that $Z \vdash s_1 = t_1$ in the graded theory that presents $\mathbb{M}$ (cf. Section 2). Then, S selects an element $(u,v) \in Z$, upon which the game reaches the new configuration $(u\sigma, v\sigma)$. The game proceeds forever unless a player cannot move. Again, any player who cannot move, loses, and infinite matches are won by D. We write $s \sim_{\mathcal{G}} t$ if D wins $\mathcal{G}_\infty^{\mathsf{syn}}$ from position (s,t).

We construct an $\bar{M}_1$-coalgebra on the set $U = \mathscr{T}_{\Sigma,\infty}/{\sim_{\mathcal{G}}}$ of infinite-depth terms modulo the winning region of D as follows. We make U into an M_0-algebra by letting depth-0 operations act by term formation. We then define the coalgebra structure $\zeta\colon U \to \bar{M}_1 U$ by $\zeta(q(t_1\sigma)) = \bar{M}_1((q \cdot \sigma)_0^*)([t_1])$ (using Kleisli star as per Notation 2.6) where $t_1\sigma$ is a top-level decomposition of an infinite-depth term, with $t_1 \in \mathscr{T}_{\Sigma,1}(X)$;

$$[-]\colon \mathscr{T}_{\Sigma,1}(X) \to M_1X = \bar{M}_1M_0X \quad \text{and} \quad q\colon \mathscr{T}_{\Sigma,\infty} \to U$$

denote canonical quotient maps. These data are well-defined.

Theorem 7.12. *The coalgebra (U,ζ) is final.*

8. Case studies

We have already seen (Example 6.9) how the standard bisimilation game arises as an instance of our generic game. We elaborate on some further examples.

Simulation equivalence. We illustrate how the infinite $(\alpha, \mathbb{M})$-behavioural equivalence game can be used to characterise simulation equivalence [20] on serial LTS. We have described the graded theory of simulation in Example 3.3.(4). Recall that it requires actions to be monotone, via the depth-1 equation $a(x+y) = a(x+y) + a(x)$. When trying to show that depth-1 terms $\sum_{i\in I} a_i(t_i)$ and $\sum_{j\in J} b_j(s_k)$ are equal, D may exploit that over join semilattices, inequalities can be expressed as equalities ($x \leq y$ iff $x + y = y$), and instead endeavour to show inequalities in both directions. By the monotonicity of actions, $\sum_{i\in I} a_i(t_i) \leq \sum_{j\in J} b_j(s_k)$ is implied by D claiming, for each i, that $t_i \leq s_j$ for some j such that $a_i = b_j$; symmetrically for $\geq$ (and by the description of the relevant graded monad as per Example 2.10.(4), this proof principle is complete). Once S challenges either a claim of the form $t_i \leq s_j$ or one of the form $t_i \geq s_j$, the direction of inequalities is fixed for the rest of the game; this corresponds to the well-known phenomenon that in the standard pebble game for similarity, S cannot switch sides after the first move. Like for bisimilarity (Example 6.9), the game can be modified to let S move first: S first picks, say, one of the terms t_i, and D responds with an s_j such that $a_i = b_j$, for which she claims $t_i \leq s_j$. Overall, the game is played on positions in $\mathscr{P}^+(X) \times \mathscr{P}^+(X)$, but if started on two states x, y of the given labelled transition systems, i.e. in a position of the form $(\{x\},\{y\})$, the game forever remains in positions where both components are singletons, and thus is effectively played on pairs of states. Summing up, we recover exactly the usual pebble game for mutual similarity. Variants such as complete, failure, or ready simulation are captured by minor modifications of the graded semantics [15].

T-structured trace equivalence. Fix a set $\mathcal{A}$ and a finitary monad T on **Set**. We are going to consider the $(\mathsf{id}, \mathbb{M}_T(\mathcal{A}))$-behavioural equivalence game on coalgebras for the functor $T(\mathcal{A}\times -)$ (cf. Example 2.5.(3)).

Notation 8.1. Fix a presentation (Σ', E') of T (i.e. an equational theory in the sense of universal algebra). We generalize the graded trace theory described in Example 2.10.(3) to a graded theory $\mathcal{T} = (\Sigma, \mathcal{E})$ for $\mathbb{M}_T(\mathcal{A})$ as follows: (Σ', E') forms the depth-0 part of $\mathcal{T}$ and, at depth-1, $\mathcal{T}$ has unary actions $a(-)$ which distribute over all operations $f \in \Sigma'$:

$$a(f(s_1, \cdots, s_{\mathsf{ar}(f)})) = f(a(s_1), \cdots, a(s_{\mathsf{ar}(f)}))$$

The arising theory $\mathcal{T}$ presents $\mathbb{M}_T(\mathcal{A})$.

Recall from Remark 6.2.(1) that since, in this setting, $M_0 = T$, a legal move for D in position $(s,t) \in TX \times TX$ is a relation Z on TX such that equality of the respective successors $(\alpha_X \cdot \gamma)_0^*(s)$ and $(\alpha_X \cdot \gamma)_0^*(t)$, viewed as (equivalence classes of) depth-1 terms, is derivable in the theory $\mathcal{T}$ under assumptions Z.

Remark 8.2. A natural question is whether there exist algorithms for deciding if a pair $(\alpha_X \cdot \gamma)_0^*(s), (\alpha_X \cdot \gamma)_0^*(t)$ sits in the congruence closure of Z. In fact, there are algorithms to check congruence closure of depth-0 terms for the powerset monad $T = \mathscr{P}_\mathsf{f}$ [11] and for certain semiring monads [10]. The idea behind those algorithms is to obtain rewrite rules from pairs in Z, and two elements are in the congruence closure if and only if they can be rewritten to the same *normal form*. Applying depth-1 equations to normal forms could potentially yield a method to check automatically whether a given pair of M_1-terms lies in the congruence closure of Z.

Finite-trace equivalence. More concretely, we examine the behavioural equivalence game for trace equivalence on finitely branching LTS (i.e. $(\mathsf{id}, \mathbb{M}_{\mathscr{P}_\mathsf{f}}(\mathcal{A}))$-semantics as per Example 3.3.(2)).

Example 8.3. Consider the following process terms representing a coalgebra γ (in a fragment of CCS):

$$p_1 \equiv a.p_1'; \quad p_2 \equiv a.p_2' + b.p_2''; \quad p_3 \equiv b.p_3',$$

where p_1', p_2', p_2'', p_3' are deadlocked. It is easy to see that $s = \{p_1, p_2\}$ and $t = \{p_2, p_3\}$ are trace equivalent: In particular, s, t have the same traces of length 1. We show that D has a winning strategy in the 1-round $(\mathsf{id}, \mathbb{M}_{\mathscr{P}_\mathsf{f}}(\mathcal{A}))$-behavioural equivalence game at (s, t). Indeed, the relation $Z := \{p_1' + p_2' = p_2',\ p_3' + p_2'' = p_2''\}$, is admissible at (s, t): We must show that equality of $(\alpha \cdot \gamma)^\#(s) = a(p_1') + a(p_2') + b(p_2'')$ and $(\alpha \cdot \gamma)^\#(t) = a(p_2') + b(p_2'') + b(p_3')$ is entailed by Z. To see this, note that

$$Z \vdash a(p_1') + a(p_2') = a(p_2') \quad \text{and} \quad Z \vdash b(p_2'') = b(p_3') + b(p_2'').$$

Moreover, the pairs $(\{p_1', p_2'\}, \{p_2'\})$ and $(\{p_3', p_2''\}, \{p_2''\})$ are both identified by M_0! (all terms are mapped to $\{\epsilon\}$ when $1 = \{\epsilon\}$). That is, Z is a winning move for D.

In general, admissible moves of D can be described via a normalisation of depth-1 terms as follows:

Proposition 8.4. *In $\mathbb{M}_{\mathscr{P}_\mathsf{f}}(\mathcal{A})$, every depth-1 term is derivably equal to one of the form $\sum_{a\in\mathcal{A}} a(t_a)$, with depth-0 terms (i.e. finite, possibly empty, sets) t_a. Over serial LTS (i.e. $T = \mathscr{P}_\mathsf{f}^+$), every depth-1 term has a normal form of the shape $\sum_{a\in B} a(t_a)$ with $B \in \mathscr{P}_\mathsf{f}^+\mathcal{A}$ (where the t_a are now finite and non-empty).*

Proposition 8.5. *Let $\rho = \sum_{a\in\mathcal{A}} a(\rho_a)$ be depth-1 terms over X in normal form, for $\rho \in \{s, t\}$. Then a relation $Z \subseteq \mathscr{P}_\mathsf{f}X \times \mathscr{P}_\mathsf{f}X$ is a legal move of D in position (s, t) iff the following conditions hold for all $a \in \mathcal{A}$, in the notation of Proposition 8.4:*

(1) $\forall x \in s_a.\ \exists t'(t' \subseteq t_a \wedge Z \vdash x \leq t')$

(2) $\forall y \in t_a.\ \exists s'(s' \subseteq s_a \wedge Z \vdash y \leq s')$

where, again, $s \leq t$ abbreviates $s + t = t$. Over serial LTS (i.e. $T = \mathscr{P}_\mathsf{f}^+$), and for normal forms $\rho = \sum_{a\in B_\rho} a(\rho_a)$, a relation $Z \subseteq \mathscr{P}_\mathsf{f}^+X \times \mathscr{P}_\mathsf{f}^+X$ is a legal move of D in position (s, t) iff $B_s = B_t$ and the above conditions hold for all $a \in B_s$.

To explain terminology, we note at this point that by the above, in particular $Z = \{(x, 0) \mid x \in X\} \cup \{(0, y) \mid y \in X\}$ is always admissible. Playing Z, D is able to move in every round, bluffing her way through the game; but this strategy does not win in general, as her bluff is called at the end (cf. Section 6). More reasonable strategies work as follows.

On the one hand, D can restrict herself to playing the bisimulation relation on the determinised transition system because the term s' (resp. t') can be taken to be exactly s_a (resp. t_a) in Condition 2 (resp. Condition 1). This form of the game may be recast as follows. Each round consists just of S playing some $a \in \mathcal{A}$ (or $a \in B_s$ in the serial case), moving to (s_a, t_a) regardless of any choice by D. In the non-serial case, the game runs until the bluff is called after the last round. In the serial case, D wins if either all rounds are played or as soon as $B_s = B_t = \emptyset$, and S wins as soon as $B_s \neq B_t$.

On the other hand, D may choose to play in a more fine-grained manner, playing one inequality $x \leq t'$ for every $x \in s_a$ and one inequality $s' \geq y$ for every $y \in t_a$. Like in the case of simulation, the direction of inequalities remains fixed after S challenges one of them, and the game can be rearranged to let S move first, picking, say, $x \in s_a$ (or symmetrically), which D answers with $t' \subseteq t_a$, reaching the new position $x \leq t'$. The game thus proceeds like the simulation game, except that D is allowed to play sets of states.

Probabilistic traces. These are treated similarly as traces in non-deterministic LTS: Every depth-1 term can be normalized into one of the form $\sum_{\mathcal{A}} p_a \cdot a(t_a)$, where $\sum_{\mathcal{A}} p_a = 1$ and the t_a are depth-0 terms. To show equality of two such normal forms $\sum_{a\in\mathcal{A}} p_a \cdot a(t_a)$ and $\sum_{a\in\mathcal{A}} q_a \cdot a(s_a)$ (arising as successors of the current configuration), D needs to have $p_a = q_a$, and then claim $t_a = s_a$, for all $a \in \mathcal{A}$. Thus, the game can be rearranged to proceed like the first version of the trace game described above: S selects $a \in \mathcal{A}$, and wins if $p_a \neq q_a$ (and the game then reaches the next configuration (t_a, s_a) without intervention by D).

Failure equivalence. Let $\gamma\colon X \to \mathscr{P}_{\mathsf{f}}(\mathcal{A} \times X)$ be an LTS. A tuple $(w, B) \in \mathcal{A}^* \times \mathscr{P}_{\mathsf{f}}(\mathcal{A})$ is a *failure pair* of a state x if there is a w-path from x to a state $x' \in X$ such that x' *fails* to perform some action $b \in B$ (the *failure set*). Two states are failure equivalent iff they have the same set of failure pairs.

The *graded theory of failure semantics* [15] extends the graded theory of traces by depth-1 constants A for each $A \in \mathscr{P}_f(\mathcal{A})$ (failure sets) and depth-1 equations $A + (A \cup B) = A$ for each $A, B \in \mathscr{P}_f(\mathcal{A})$ (failure sets are downwards closed). The resulting graded monad [15] has $M_0X = \mathscr{P}_{\mathsf{f}}X$ and $M_1X = \mathscr{P}^{\downarrow}_{\mathsf{f}}(\mathcal{A} \times X + \mathscr{P}_{\mathsf{f}}\mathcal{A})$, where $\mathscr{P}_{\mathsf{f}}\mathcal{A}$ is ordered by inclusion, $\mathcal{A} \times X$ carries the discrete order, and $\mathscr{P}^{\downarrow}_{\mathsf{f}}$ denotes the finitary downwards-closed powerset. It is clear that M_1 still preserves monos since we have only expanded the theory of traces by constants. The game in general is then described similarly as the one for plain traces above; the key difference is that now S is able to challenge whether a pair of failure sets are matched up to downwards closure.

Example 8.6. Consider the following process terms with $\mathcal{A} = \{a, b, c\}$: $p_1 \equiv a.\mathbf{0}$, $p_2 \equiv a.\mathbf{0} + b.\mathbf{0}$, $p_3 \equiv b.\mathbf{0}$.

Clearly, the states $s = \{p_1, p_2, p_3\}$ and $t = \{p_1, p_3\}$ in the pre-determinized system are failure equivalent. To reason this through our game $\mathcal{G}_\infty(\gamma)$, Duplicator starts with the relation $Z = \{(\mathbf{0}, \mathbf{0})\}$. From Z, we derive

$$\begin{aligned}(\alpha_X \cdot \gamma)^*_0 s &= a(\mathbf{0}) + b(\mathbf{0}) + {\downarrow}\{b, c\} + {\downarrow}\{c\} + {\downarrow}\{a, c\}\\ &= a(\mathbf{0}) + b(\mathbf{0}) + {\downarrow}\{b, c\} + {\downarrow}\{c\} + {\downarrow}\{c\} + {\downarrow}\{a, c\}\\ &= a(\mathbf{0}) + b(\mathbf{0}) + {\downarrow}\{b, c\} + {\downarrow}\{a, c\} = (\alpha_X \cdot \gamma)^*_0 t.\end{aligned}$$

Thus Z is admissible at (s, t) and the game position advances to $(\mathbf{0}, \mathbf{0})$ from where Duplicator has a winning strategy.

9. Conclusions and Future Work

We have shown how to extract characteristic games for a given graded behavioural equivalence, such as similarity, trace equivalence, or probablistic trace equivalence, from the underlying graded monad, effectively letting Spoiler and Duplicator play out an equational proof. The method requires only fairly mild assumptions on the graded monad; specifically, the extension of the first level of the graded monad to algebras for the zero-th level needs to preserve monomorphisms. This condition is not completely for free but appears to be unproblematic in typical application scenarios. In case the zero-th level of the graded monad preserves the terminal object (i.e. the singleton set), it turns out that the induced graded behavioural equivalence can be recast as standard coalgebraic behavioural equivalence in a category of Eilenberg-Moore algebras, and is then characterized by an infinite version of the generic equivalence game. A promising direction for future work is to develop the generic algorithmics and complexity theory of the infinite equivalence game, which has computational content via the implied fixpoint characterization. Moreover, we will extend the framework to cover further notions of process comparison such as behavioural preorders [18] and, via a graded version of quantitative algebra [33], behavioural metrics.

Appendix A. Omitted Proofs and Details

A.1. **Proofs for Section 6.**

Lemma A.1. *Let $\mathbb{M}$ be a depth-1 graded monad and let X, Y be sets. Then for every function $f\colon X \to M_k Y$ we have*

$$\overline{M}_1(f_0^*) = f_1^*.$$

Proof. It follows by equational reasoning using the associative laws of $\mathbb{M}$ and naturality of $\mu^{0,n}$ that f_n^* is a homomorphism of M_0-algebras from $(M_n X, \mu^{0,n})$ to $(M_{n+m}X, \mu^{0,(n+m)})$. For the homomorphism $f_0^*\colon M_0X \to M_kX$ there exists, by canonicity of (M_0X, M_1X) (see Example 4.8), a unique homomorphism $h\colon M_1X \to M_{k+1}X$ making (f_0^*, h) a homomorphism of M_1-algebras. Thus, it is enough to show that (f_0^*, f_1^*) is an M_1-algebra homomorphism. Indeed, it will then follow that f_1^* is the 1-part of the free extension of f_0^* to an M_0-algebra homomorphism, i.e. $\overline{M}_1(f_0^*) = f_1^*$, as required. We conclude the proof by showing that (f_0^*, f_1^*) satisfies the remaining interchange law stating that

$$\mu^{1,k}_{M_{1+k}1} \cdot M_1(f_0^*) = f_1^* \cdot \mu^{1,0}_X$$

making it an M_1-algebra homomorphism:

$$\begin{aligned}
\mu^{1,k}_{M_{1+k}1} \cdot M_1(f_0^*) &= (f_0^*)_1^* && \text{def. of } (-)_1^* \\
&= (f_0^* \cdot \mathsf{id}_{M_0X})_1^* \\
&= f_1^* \cdot (\mathsf{id}_{M_0X})_1^* && \text{by (2.6)} \\
&= f_1^* \cdot \mu^{1,0}_X && \text{def. of } (-)_1^*.
\end{aligned}$$

□

Example A.2. Let $(\alpha, \mathbb{M})$ be a depth-1 graded semantics on G-coalgebras and let γ be a G-coalgebras. We have

$$(k \in \omega) \qquad \overline{M}_1(\gamma^{(k)})_0^* = (\gamma^{(k)})_1^*$$

by instantiation of Lemma A.1 to the map $\gamma^{(k)}\colon X \to M_kX$.

Proof of Theorem 6.6.

Remark A.3. Whenever $c\colon X \twoheadrightarrow C$ is the coequalizer of the kernel pair $p, q\colon Z \to X$ of a morphism $f\colon X \to Y$, then there is a unique morphism $m\colon C \to Y$ such that $m \cdot c = f$. Recall (e.g. from the proof of Borceaux [13, Thm. 2.1.3]) that if $\mathscr{C}$ is a regular category, then m is monic; in particular, this holds when $\mathscr{C}$ is the category of algebras for a monad on **Set**.

Proof (Theorem 6.6). Let us denote the *winning region* for D in $\mathcal{G}_n(\gamma)$ by $\mathsf{Win}_n(\mathrm{D})$ (leaving the coalgebra (X, γ) implicit in the notation). We are going to show that $(s,t) \in \mathsf{Win}_n(D)$ if and only if $(\gamma^{(n)})_0^*$ merges s and t by induction on $n \in \omega$. For $n = 0$, we must show that $M_0!(s) = M_0!(t)$ if and only if $(\gamma^{(0)})_0^*$ merges s and t. In fact, we show that $M_0! = (\gamma^{(0)})_0^*$. First note that the following diagram commutes due to the unit law (2.1) of $\mathbb{M}$ (instantiated to $n = 0$) and naturality of $\mu^{0,0}$:

$$\begin{array}{ccccc}
M_0X & \xrightarrow{M_0\eta_X} & M_0M_0X & \xrightarrow{M_0M_0!} & M_0M_01 \\
& \searrow^{\mathsf{id}_{M_0X}} & \downarrow{\scriptstyle \mu^{0,0}_X} & & \downarrow{\scriptstyle \mu^{0,0}_1} \\
& & M_0X & \xrightarrow{M_0!} & M_01
\end{array}$$

Then unfold the definition of $(\gamma^{(0)})_0^*$ and compute

$$\begin{aligned}
(\gamma^{(0)})_0^* &= \mu_1^{0,0} \cdot M_0\gamma^{(0)} \\
&= \mu_1^{0,0} \cdot M_0(M_0! \cdot \eta_X) \\
&= \mu_1^{0,0} \cdot M_0M_0! \cdot M_0\eta_X \\
&= M_0!,
\end{aligned}$$

where the last step uses that the outside of the previous diagram commutes.

Now, inductively assume that $(s,t) \in \mathsf{Win}_k(D)$ if and only if $(\gamma^{(k)})_0^*$ merges s and t. We proceed to show that the equivalence carries through to $n = k + 1$:

($\Leftarrow$) Let $(s,t) \in M_0X \times M_0X$ such that $(\gamma^{(k+1)})^*$ merges s and t be given; we must show that D has a winning strategy in the $(k+1)$-round $\mathcal{S}$-behavioural equivalence game at (s,t). Define

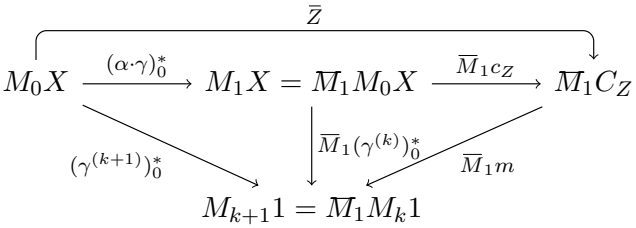
$$Z := \mathsf{ker}((\gamma^{(k)})^*_0).$$

That is, Z is the pullback, computed in **Set**, as in the diagram

$$\begin{array}{ccc} Z & \xrightarrow{r} & M_0X \\ {\scriptstyle \ell}\downarrow & & \downarrow{\scriptstyle (\gamma^{(k)})^*_0} \\ M_0X & \xrightarrow{(\gamma^{(k)})^*_0} & M_k1 \end{array}$$

It is clear that (the relation induced by) Z is a winning move for D at (s,t) provided that it is admissible. Indeed, for every $z \in Z$ we have that $(\gamma^{(k)})^*_0$ merges $\ell(z)$ and $r(z)$ by construction; so, by induction, $Z \subseteq \mathsf{Win}_k(D)$, as claimed. Thus, to conclude the proof of *'if'*, it suffices to show that Z is admissible at (s,t), i.e. we want to show that the homomorphism $Z\colon M_0X \to \overline{M}_1C_Z$ from (6.1) merges s and t.

Let $m\colon C_Z \to M_k1$ be the unique (mono-)morphism such that $m \cdot c_Z = (\gamma^{(k)})^*_0$ according to Remark A.3, and consider the following diagram of M_0-algebra homomorphisms:



The above diagram commutes: the upper part is the definition of Z, the right hand triangle commutes since $m \cdot c_Z = (\gamma^{(k)})^*_0$, and the left-hand triangle commutes by the following:

$$\begin{aligned} \overline{M}_1(\gamma^{(k)})^*_0 \cdot (\alpha\cdot\gamma)^*_0 &= (\gamma^{(k)})^*_1 \cdot (\alpha\cdot\gamma)^*_0 && \text{by Lemma A.1} \\ &= ((\gamma^{(k)})^*_1 \cdot (\alpha\cdot\gamma))^*_0 && \text{by Eq. (2.6)} \\ &= (\gamma^{(k+1)})^*_0 && \text{def. of } \gamma^{(k+1)}. \end{aligned}$$

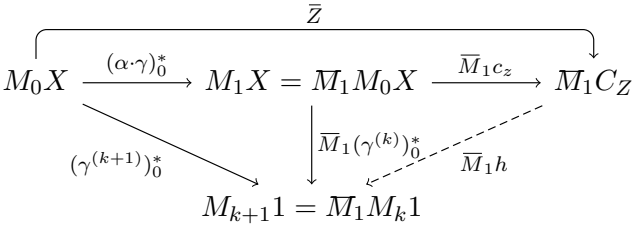
Thus

$$\overline{M}_1m \cdot Z(s) = (\gamma^{(k+1)})^*_0(s) = (\gamma^{(k+1)})^*_0(t) = \overline{M}_1m \cdot Z(t).$$

Since m is a monomorphism and $\overline{M}_1$ preserves monomorphisms, we conclude that $Z(s) = Z(t)$ holds. Hence Z is admissible at (s,t), as desired.

($\Rightarrow$) Suppose that $(s,t) \in \mathsf{Win}_{k+1}(\mathbb{D})$ so that there exists $Z \subseteq \mathsf{Win}_k(\mathbb{D})$ which is admissible at (s,t); we proceed to show that $(\gamma^{(k+1)})^*_0$ merges s and t, as required. By admissibility of Z, we know that $Z\colon M_0X \to \overline{M}_1C_Z$ from (6.1) merges s and t.

Next we see that $(\gamma^{(k)})^*_0\colon M_0X \to M_k1$ coequalizes the pair $\ell^*_0, r^*_0\colon M_Z \to M_0X$. Indeed, since $Z \subseteq \mathsf{Win}_k(D)$, we have by induction that $(\gamma^{(k)})^*_0(\ell^*_0(z)) = (\gamma^{(k)})^*_0(r^*_0(z))$ for all $z \in Z$. Thus, by the universal property of the coequalizer $c_Z\colon M_0X \to C_Z$ we obtain a unique homomorphism $h\colon C_Z \to M_k1$ such that $h \cdot c_Z = (\gamma^{(k)})^*_0$. Therefore the right-hand triangle of the following diagram of M_0-algebra homomorphisms commutes:



Note that the upper part commutes by the definition (6.1) of Z. The left-hand triangle commutes by Lemma A.1 as before. We conclude that $(\gamma^{(k+1)})^*_0$ merges s and t, as desired, since so does Z. □

A.2. **Proofs for Section 7.**

Details for Remark 7.2. We will prove that for every accessible graded monad $\mathbb{M}$ on a locally presentable category, the category $\mathsf{Alg}_n(\mathbb{M})$, $n \leq \omega$, is locally presentable. Furthermore, we show that the functor $\overline{M}_1 \colon \mathsf{Alg}_0(\mathbb{M}) \to \mathsf{Alg}_0(\mathbb{M})$ is then accessible and therefore has a final coalgebra.

Our proof makes use of several facts from the theory of accessible and locally presentable categories which we tersely recall from Adámek and Rosický's book [6].

As before, we fix a regular cardinal κ. A diagram $D \colon \mathscr{D} \to \mathscr{C}$ is *κ-filtered* if its diagram scheme $\mathscr{D}$ is a *κ-filtered* category, that is a category in which every subcategory with less than κ objects and morphisms has a cocone. (In particular, $\mathscr{D}$ is then non-empty since a cocone for the empty subcategory is some object of $\mathscr{D}$.) A functor $F \colon \mathscr{C} \to \mathscr{D}$ is *κ-accessible* if it preserves κ-filtered colimits, and a monad is κ-accessible if so is its underlying functor. An object X of a category $\mathscr{C}$ is *κ-presentable* if its hom-functor $\mathscr{C}(X, -) \colon \mathscr{C} \to \mathbf{Set}$ is κ-accessible.

Remark A.4. We shall need the following facts about κ-presentable objects and left adjoints.

(1) A left adjoint $L \colon \mathscr{C} \to \mathscr{D}$ with a κ-accessible right adjoint R preserves κ-presentable objects. Indeed, for every κ-presentable object C in $\mathscr{C}$ and every κ-filtered diagram D in $\mathscr{D}$ we have the following chain of natural isomorphisms

$$\begin{aligned} \mathscr{D}(LC, \mathsf{colim}\, D) &\cong \mathscr{C}(C, R(\mathsf{colim}\, D)) \\ &\cong \mathscr{C}(C, \mathsf{colim}\, RD) \\ &\cong \mathsf{colim}\, \mathscr{C}(C, RD) \\ &\cong \mathsf{colim}\, \mathscr{D}(LC, D). \end{aligned}$$

(2) Recall (e.g. [4, Def. 7.74]) that an epimorphism e is *extremal* if whenever $e = m \cdot f$ for a monomorphism m, then m is an isomorphism. Let $L \dashv R \colon \mathscr{D} \to \mathscr{C}$ be an adjunction where the right adjoint R is faithful and reflects isomorphisms. Then the adjoint transpose of every extremal epimorphism $e \colon C \twoheadrightarrow RD$ is an extremal epimorphism, too.

Indeed, to see that the adjoint transpose $\bar{e} \colon LC \to D$ is epic take a pair of morphisms $f, g \colon D \to D'$ such that $f \cdot \bar{e} = g \cdot \bar{e}$. Then $Uf \cdot e = Ug \cdot e$ by adjoint transposition, whence $Uf = Ug$ since e is epic, and therefore $f = g$ since U is faithful.

Now suppose that $\bar{e} = m \cdot f$ for some monomorphism m in $\mathscr{D}$. Then $e = Um \cdot \bar{f}$ in $\mathscr{C}$, where Um is monic since right adjoints preserve monos. Thus, Um is an isomorphism in $\mathscr{C}$, whence m is an isomorphism in $\mathscr{D}$ since U reflects isomorphisms.

A category $\mathscr{C}$ is *κ-accessible* if it has κ-filtered colimits and a set of κ-presentable objects such that every object of $\mathscr{C}$ is a κ-filtered colimit of object from that set. The category $\mathscr{C}$ is locally κ-presentable if it is locally κ-accessible and cocomplete. We say that a category is *accessible* (*locally presentable*) if it is κ-accessible (locally κ-presentable) for some regular cardinal κ.

Remark A.5. We now collect several facts used in the proof of Theorem A.6 below.

(1) A category is locally κ-presentable iff is is κ-accessible and complete [6, Cor. 2.47].

(2) A *strong generator* in a cocomplete category $\mathscr{C}$ is a set $\mathscr{G}$ of objects such that every object is an extremal quotient of a coproduct of object from $\mathscr{G}$. A cocomplete category is locally κ-presentable iff it has a strong generator formed by κ-presentable objects [6, Thm. 1.20].

(3) For a κ-accessible monad T on a locally κ-presentable category $\mathscr{C}$, the category $\mathsf{Alg}(T)$ of all Eilenberg-Moore algebras is locally κ-presentable [6, Thm. & Rem. 2.78].

(4) For every locally κ-presentable category $\mathscr{C}$ the product category $\mathscr{C} \times \mathscr{C}$ is locally κ-presentable. Indeed, first $\mathscr{C} \times \mathscr{C}$ is κ-accessible [6, Prop. 2.67]. In addition, the product is (co)complete since so is $\mathscr{C}$ and (co)limits are formed componentwise in the product category. Moreover, the product projection functors are clearly κ-accessible.

(5) Let $\mathscr{C}$ and $\mathscr{K}$ be locally κ-presentable, and let $F, G \colon \mathscr{K} \to \mathscr{C}$ be κ-accessible. The *inserter category* of F, G has objects (K, f) where K is an object of $\mathscr{K}$ and $f \colon FK \to GK$ is a morphism of $\mathscr{C}$; morphisms $h \colon (K, f) \to (K', f')$ are morphisms $h \colon K \to K'$ of $\mathscr{K}$ such that $f' \cdot Fk = Gk \cdot f$. The inserter category is λ-accessible for some regular cardinal $\lambda \geq \kappa$ [6, Thm. 2.72]. (Note that, in general, $\lambda \neq \kappa$.)

(6) Let $\mathscr{C}$ and $\mathscr{K}$ be κ-accessible, and let $F, G \colon \mathscr{K} \to \mathscr{C}$ be κ-accessible functors. The *equifier* of a pair $\alpha, \beta \colon F \to G$ of natural transformations is the full subcategory of $\mathscr{K}$ given by all objects K such that $\alpha_K = \beta_K$. This category is κ-accessible and its inclusion functor into $\mathscr{K}$ is κ-accessible.

The same holds, more generally, for any set of pairs $F^i, G^i\colon \mathscr{K} \to \mathscr{C}$ $(i \in I)$ and any set of pairs $\alpha^i, \beta^i\colon F^i \to G^i$ [6, Lem. 2.76].

We say that a graded monad $\mathbb{M}$ on $\mathscr{C}$ is *κ-accessible* if every of its endofunctors M_n $(n \in \omega)$ is κ-accessible.

Theorem A.6. *Let $\mathbb{M}$ be a κ-accessible graded monad on a locally κ-presentable category $\mathscr{C}$. Then the category $\mathsf{Alg}_n(\mathbb{M})$ is locally κ-presentable for every $n \leq \omega$.*

Remark A.7. The proof is essentially a more involved variation on the proof of the result on categories $\mathsf{Alg}(T)$ of Eilenberg-Moore algebras mentioned in Remark A.5.(3). We provide the details for the convenience of the reader.

Proof. For $n = 0$ we are done since $\mathsf{Alg}_0(\mathbb{M})$ is simply the Eilenberg-Moore category for the κ-accessible monad M_0. We now give an explicit proof for $n = 1$; the general case is then an easy exercise.

(1) Let $\mathscr{K} = \mathsf{Alg}_0(\mathbb{M}) \times \mathsf{Alg}_0(\mathbb{M})$ be the product of the Eilenberg-Moore category for the monad M_0 with itself. Then $\mathscr{K}$ is locally κ-presentable by Remark A.5.(4).

(2) Let $P_0, P_1\colon \mathscr{K} \to \mathscr{C}$ be the functors obtained by composing the product projections with the forgetful functor $U_0\colon \mathsf{Alg}_0(\mathbb{M}) \to \mathscr{C}$. By Remark A.5.(5), the inserter category $\mathscr{I}$ of the pair M_1P_0, P_1 of κ-accessible functors is λ-accessible for some regular cardinal $\lambda \geq \kappa$.

(3) Note that objects of the inserter category $\mathscr{I}$ are 5-tuples $A = (A_0, A_1, (a^{i,j})_{i+j\leq 1})$, where $a^{i,j}\colon M_iA_j \to A_{i+j}$ is a morphism of $\mathscr{C}$ (i.e. the same data as objects of $\mathsf{Alg}_1(\mathbb{M})$) such that $(A_0, a^{0,0})$ and $(A_1, a^{0,1})$ are Eilenberg-Moore algebras for the monad M_0 (the remaining axioms, homorphy and coequalization, of M_1-algebras need not hold). Let $V\colon \mathscr{I} \to \mathscr{K}$ be the functor forgetting the structure morphism $a^{1,0}$.

(4) We prove that V reflects isomorphisms. First note that morphisms in $\mathscr{I}$ from $(A_0, A_1, (a^{i,j})_{i+j\leq 1})$ to $(B_0, B_1, (b^{i,j})_{i+j\leq 1})$ are pairs $h_i\colon A_i \to B_i$, $i = 0, 1$, of M_0-algebra homomorphisms such that, in addition, we have $h_1 \cdot a^{1,0} = b^{1,0} \cdot M_1h_0$. Now given such a pair such that $V(h_0, h_1)$ is an isomorphism in $\mathscr{K}$, that means that h_i is an isomorphism of M_0-algebras for $i = 0, 1$, we need to show that the inverses h_i^{-1} form a morphism in $\mathscr{I}$. To this end we make use of the following diagram:

$$\begin{array}{ccccc}
 & & M_1B_0 & \xrightarrow{b^{1,0}} & B_1 & & \\
 & {\scriptstyle M_1h_0^{-1}}\downarrow & & & \downarrow{\scriptstyle h_1^{-1}} & & \\
{\scriptstyle M_1\mathsf{id}} & M_1A_0 & \xrightarrow{a^{1,0}} & A_1 & & {\scriptstyle \mathsf{id}} \\
 & {\scriptstyle M_1h_0}\downarrow & & & \downarrow{\scriptstyle h_1} & & \\
 & M_1B_0 & \xrightarrow{b^{1,0}} & B_1 & &
\end{array}$$

Indeed, the outside, left- and right-hand parts and the lower square commute. Thus so does the desired upper square when postcomposed by the isomorphism h_1. Thus, this square commutes.

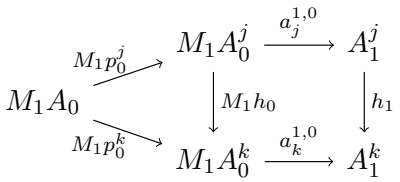
(5) We now prove that V creates limits and κ-filtered colimits. Observe first that limits and κ-filtered colimits of M_0-algebras are formed at the level of their carrier objects since the forgetful functor $U_0\colon \mathsf{Alg}_0(\mathbb{M}) \to \mathscr{C}$ creates limits and κ-filtered colimits (the latter since M_0 preserves them). Further note that all limits and colimits in $\mathscr{K}$ are formed componentwise.

For limits, given a diagram of objects A^j $(j \in J)$ in $\mathscr{I}$, let A be its limit in $\mathscr{K}$ with the limit projections $p^j\colon A \to A^j$. Then we obtain a unique morphism $a^{1,0}\colon M_1A_0 \to A_1$ such that every p^j $(j \in J)$ is a morphism in $\mathscr{I}$, i.e. the following squares commute (throughout we put the index j of the structure morphisms of A^j in the subscript):

$$\begin{array}{ccc}
M_1A_0 & \overset{a^{1,0}}{\dashrightarrow} & A_1 \\
{\scriptstyle M_1p_0^j}\downarrow & & \downarrow{\scriptstyle p_1^j} \\
M_1A_0^j & \xrightarrow{a_j^{1,0}} & A_1^j
\end{array} \qquad \text{for every } j \in J.$$

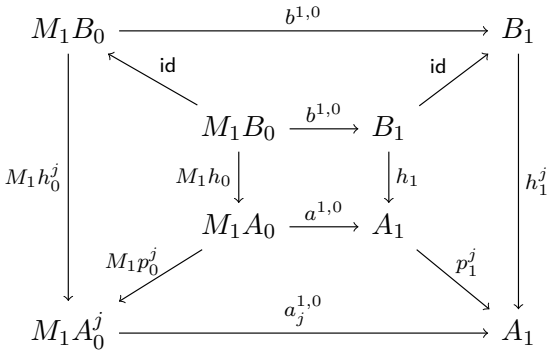
Indeed, the morphisms $a_j^{1,0} \cdot M_1p_0^j$ form a cone (in $\mathscr{C}$) on the diagram yielding the component A_1 of the colimit A: for every morphism $h = (h_0, h_1)\colon A^j \to A^k$ in the given diagram scheme we have a

commutative diagram



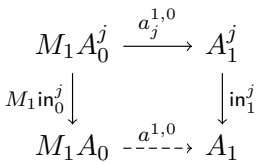
Thus, A carries a unique structure of an $\mathscr{I}$-object such that the limit projections p^j are morphisms of $\mathscr{I}$.

We still need to prove that A is the limit of the A^j in $\mathscr{I}$. Given any cone $h^j\colon B \to A^j$ $(j \in J)$ in $\mathscr{I}$, we know that there is a unique morphism $h\colon B \to A$ of $\mathscr{K}$ such that $p^j \cdot h = h^j$ for every $j \in J$. It suffices to show that h is a morphism in $\mathscr{I}$. For this consider the following diagram:



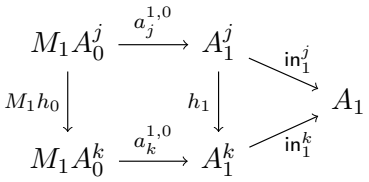
The outside commutes since h^j is a morphism of $\mathscr{I}$. The left- and right-hand parts commute by the definition of h, the lower part commutes by the definition of $a^{1,0}$ and the upper part trivially does. Thus, the middle square commutes when postcomposed by every limit projection p^j_1, whence this square commutes, as desired.

For κ-filtered colimits, given a κ-filtered diagram of M_1-algebras A^j $(j \in J)$, let A be its colimit in $\mathscr{I}$ with colimit injections $\mathsf{in}^j\colon A^j \to A$. We use that M_1 preserves the colimit in the first component; that is, we have a κ-filtered colimit with the injections $M_1\mathsf{in}^j_0\colon M_1A^j_0 \to M_1A_0$ $(j \in J)$ in $\mathsf{Alg}_0(\mathbb{M})$ (whence in $\mathscr{C}$). Then we obtain a unique algebra structure $a^{1,0}\colon M_1A_0 \to A_1$ such that every in^j $(j \in J)$ is a morphism in $\mathscr{I}$, i.e. the following squares commute:



for every $j \in J$.

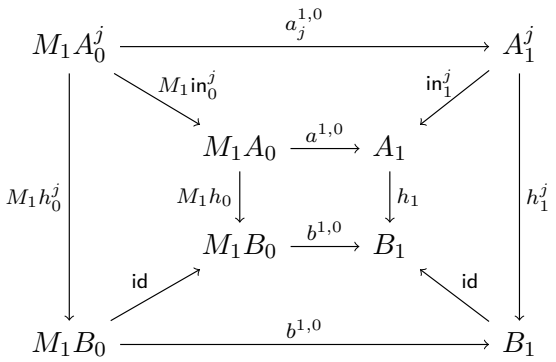
Indeed, the morphisms $\mathsf{in}^j_1 \cdot a^{1,0}_j\colon M_1A^j_0 \to A_1$ form a cocone of the above diagram formed by the M_0-algebras $M_1A^j_0$: for every morphism $h = (h_0, h_1)\colon A^j \to A^k$ in the diagram scheme we have a commutative diagram



Thus, A carries a unique structure of an $\mathscr{I}$-object such that the colimit injections in^j are morphisms of $\mathscr{I}$.

We still need to prove that A is the colimit of the A^j in $\mathscr{I}$. Given any cocone $h^j\colon A^j \to B$ $(j \in J)$ in $\mathscr{I}$ we know that there is a unique morphism $h\colon A \to B$ of $\mathscr{K}$ such that $h \cdot \mathsf{in}^j = h^j$ for every $j \in J$.

It suffices to show that h is a morphism in $\mathscr{I}$. For this consider the following diagram:



The outside commutes since h^j is a morphism in $\mathscr{I}$. The left- and right-hand parts commmute by the definition of h, the upper part commutes by the definition of $a^{1,0}$ and the lower part trivially does. Thus, the inner square commutes when precomposed by every colimit injection $M_1\mathsf{in}_0^j$, whence it commutes, as desired.

(6) It follows that $\mathscr{I}$ is complete, and therefore it is locally presentable by Remark A.5.(1). Moreover, by the Adjoint Functor Theorem [6, Thm. 1.66], V has a left adjoint $L\colon \mathscr{K} \to \mathscr{I}$.

From Remark A.4.(1) we obtain that for every κ-presentable object in $\mathscr{K}$, i.e. a pair A formed by κ-presentable objects $(A_i, a^{0,i})$, $i = 0, 1$, the object LA is κ-presentable in $\mathscr{I}$. We show that the these κ-presentable objects LA form a strong generator in $\mathscr{I}$. By Remark A.5.(2), it follows that $\mathscr{I}$ is locally κ-presentable. Given an object A of $\mathscr{I}$ we know that VA is an extremal quotient of a coproduct of κ-presentable objects in $\mathscr{K}$, reprensented, say, by the extremal epimorphism

$$e\colon \coprod_{j\in J} A_j \twoheadrightarrow VA$$

for some set J. Using Remark A.4.(1), that the left-adjoint L preserves coproducts and that V reflects isomorphisms and is clearly faithful we thus see that the adjoint transpose

$$\bar{e}\colon \coprod_{j\in J} LA_j \twoheadrightarrow A$$

exhibits A as an extremal quotient of the coproduct of the κ-presentable objects LA_j.

(7) Now let $Q_i\colon \mathscr{I} \to \mathscr{C}$, $i = 0, 1$, be the obvious forgetful projection functors. We have natural transformations $\varphi^{i,j}\colon M_iQ_j \to Q_{i+j}$ for $i + j \leq 1$ with components $\varphi^{i,j}_A = a^{i,j}$. The missing laws of an M_1-algebra (homomorphy and coequalization) can be expressed by equifiers of the following pairs of natural transformations

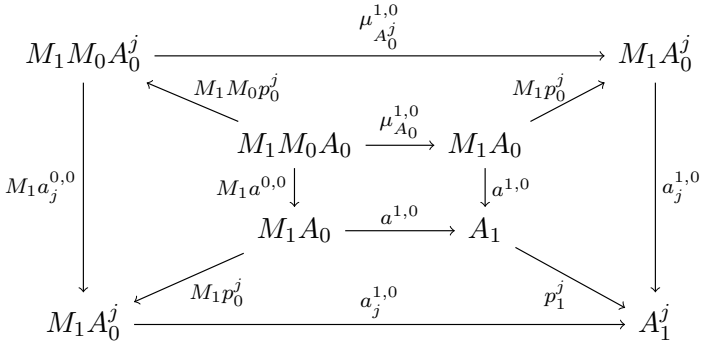
$$\varphi^{1,0} \cdot \mu^{1,0}Q_0, \varphi^{1,0} \cdot M_1\varphi^{0,0}\colon M_1M_0Q_0 \to Q_1,$$
$$\varphi^{1,0} \cdot \mu^{1,0}Q_0, \varphi^{1,0} \cdot M_0\varphi^{1,0}\colon M_0M_1Q_0 \to Q_1.$$

Consequently, $\mathsf{Alg}_1(\mathbb{M})$ is κ-accessible and the inclusion functor $I\colon \mathsf{Alg}_1(\mathbb{M}) \hookrightarrow \mathscr{I}$ is κ-accessible by Remark A.5.(6).

(8) To see that $\mathsf{Alg}_1(\mathbb{M})$ is locall κ-presentable, we prove that $\mathsf{Alg}_1(\mathbb{M})$ is closed under limits and κ-filtered colimits in $\mathscr{I}$. This is a slight variation of the standard argument that the Eilenberg-Moore algebras for a κ-accessible monad T is closed under limits and κ-filtered colimits.

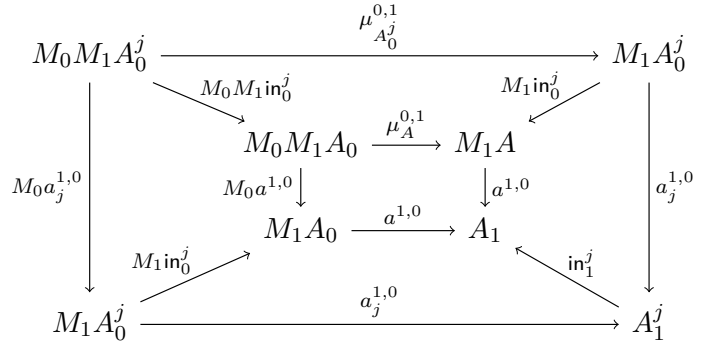
For limits, given a diagram of M_1-algebras A^j ($j \in J$), let A be its limit in $\mathscr{I}$ with limit projections $p^j\colon A \to A^j$. We show that A is an M_1-algebra, that means that it satisfies homomorphy and coequalization. For the latter consider the following diagram (structure morphisms of A^j have their

index as a subscript, e.g. $a_j^{1,0}\colon M_1A_0^j \to A_1^j$):



The outside commutes due to coequalization of A^j and the four inner trapezoids by the naturality of $\mu^{1,0}$ and since $p^j = (p_0^j, p_1^j)$ is a morphism in $\mathscr{I}$. Thus, the desired middle square commutes when postcomposed by the projection p_1^j. Since limits in $\mathscr{I}$ are formed componentwise, the p_1^j $(j \in J)$ form a limit cone whence a jointly monic family. This, implies that the desired inner square commutes. That A satisfies homomorphy is shown analogogously.

For κ-filtered colimits, given a κ-filtered diagram of M_1-algebras A^j $(j \in J)$, let A be its colimit in $\mathscr{I}$ with colimit injections $\mathsf{in}^j\colon A^j \to A$. We show that A is an M_1-algebra, that means that it satisfies homomorphy and coequalization. For the former, observe first that since M_0M_1 is κ-accessible and colimits in $\mathscr{I}$ are formed componentwise we have a colimit cocone $M_0M_1\mathsf{in}_0^j\colon A_0^j \to A_0$. Now consider the following diagram (again we put the index of A^j in the subscript of structure morphisms)

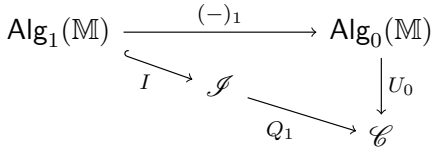


The outside commutes since A^j is an M_1-algebra, and the four inner trapezoids commute due to the naturality of $\mu^{0,1}$ and the fact that in^j is a morphism in $\mathscr{I}$. Thus, the desired inner square commutes when precomposed by every colimit injection $M_0M_1\mathsf{in}_0^j$, whence it commutes as desired. That A satisfies coequalization is shown analogously

(9) The desired result now follows from the Reflection Theorem [6, Thm. 2.48]. □

Corollary A.1. *The functor* $(-)_1\colon \mathsf{Alg}_1(\mathbb{M}) \to \mathsf{Alg}_0(\mathbb{M})$ *taking* 1*-parts is* κ*-accessible.*

Indeed, this follows from the proof of Theorem A.6 since we have the commutative diagram of functors



where the functor Q_1I preserves and the forgetful functor U_0 creates κ-filtered colimits.

Corollary A.2. *The functor* $\overline{M}_1\colon \mathsf{Alg}_0(\mathbb{M}) \to \mathsf{Alg}_0(\mathbb{M})$ *has a final coalgebra.*

Indeed, since $\overline{M}_1 = (-)_1 \cdot E$, where E is a left adjoint (see (5.1)) we see that $\overline{M}_1$ is a κ-accessible endofunctor on the locally κ-presentable category $\mathsf{Alg}_0(\mathbb{M})$. It follows that the category of all coalgebra for $\overline{M}_1$ is locally presentable [6, Exercise 2j]. (Though note that the index of presentability may not be κ.) In particular, this category is complete and therefore has a terminal object.

Proof of Theorem 7.5.

Remark A.8. Suppose that $\mathscr{C}$ is a category with a (regular epi, mono) factorization system (e.g. a regular category [13, Def. 2.1.1]); that means that every morphism f of $\mathscr{C}$ has a factorization $f = m \cdot e$ into a regular epimorphism e followed by a monomorphism m. Moreover, the following unique diagonal fill-in property holds: whenever we have $m \cdot f = g \cdot e$ for a regular epimorphism e and a monomorphism m, then there exists a unique diagonal d such that $m \cdot d = g$ and $d \cdot e = f$:

$$\begin{array}{ccc} A & \xrightarrow{\ e\ } & B \\ {\scriptstyle f}\downarrow & \overset{d}{\swarrow} & \downarrow{\scriptstyle g} \\ C & \xrightarrow{\ m\ } & D \end{array}$$

Now suppose further that $F\colon \mathscr{C} \to \mathscr{C}$ preserves monomorphisms. Then every coalgebra morphism $h\colon (A,\gamma) \to (B,\delta)$ can be factorized into a coalgebra morphism carried by a regular epi and one carried by a monomorphism in $\mathscr{C}$. Indeed, take the factorization h into a regular epimorphism $e\colon A \twoheadrightarrow C$ followed by a monomorphism $m\colon C \rightarrowtail B$ in $\mathscr{C}$, and observe that the unique diagonal fill-in property yields a unique coalgebra structure $\alpha\colon C \to FC$ such that e and m are coalgebra morphisms (here one uses that Fm is a monomorphism):

$$\begin{array}{ccc} A & \xrightarrow{\ \gamma\ } & FA \\ {\scriptstyle e}\downarrow & & \downarrow{\scriptstyle Fe} \\ C & \overset{\alpha}{\dashrightarrow} & FC \\ {\scriptstyle m}\downarrow & & \downarrow{\scriptstyle Fm} \\ B & \xrightarrow{\ \delta\ } & FB \end{array} \qquad h,\ Fh$$

Towards the proof of Theorem 7.5, we generalize the game to be played on any $\overline{M}_1$-coalgebra (A,γ). Positions of D then are pairs in $A \times A$; positions of S are relations $Z \subseteq A \times A$. A move from $(x,y) \in A \times A$ to such a Z is admissible for D if the relations in Z entail equality of $\gamma(x)$ and $\gamma(y)$; in categorical formulation, this means that $\overline{M}_1 c_Z(\gamma(x)) = \overline{M}_1 c_Z(\gamma(y))$ where $c_Z\colon A \to C_Z$ is the coequalizer of the pair $\ell_0^*, r_0^*\colon M_0 Z \to A$ for the projections $\ell, r\colon Z \to A$. Again, S just picks from Z; any player who cannot move, loses, and infinite matches are won by D. We then claim, generalizing the claim of the theorem, that $x, y \in A$ are behaviourally equivalent iff D wins (x,y).

Proof (Theorem 7.5). ($\Leftarrow$) We denote the winning region of D by $\sim_{\mathcal{G}}$. It is easy to see that $\sim_{\mathcal{G}}$ is a congruence on the M_0-algebra A (this is similar as in the proof of Theorem 7.12), so we have a unique M_0-algebra structure on $A/{\sim_{\mathcal{G}}}$ such that the quotient map $e\colon A \twoheadrightarrow A/{\sim_{\mathcal{G}}}$ is an M_0-algebra homomorphism. We write elements of $A/{\sim_{\mathcal{G}}}$ in the form $[x]_D$ for $x \in A$. We shall define a coalgebra structure $\gamma_D\colon A/{\sim_{\mathcal{G}}} \to \overline{M}_1(A/{\sim_{\mathcal{G}}})$ such that the square below commutes in $\mathsf{Alg}_0(\mathbb{M})$:

$$\begin{array}{ccc} A & \xrightarrow{\ \gamma\ } & \overline{M}_1 A \\ {\scriptstyle e}\downarrow & & \downarrow{\scriptstyle \overline{M}_1 e} \\ A/{\sim_{\mathcal{G}}} & \overset{\gamma_D}{\dashrightarrow} & \overline{M}_1(A/{\sim_{\mathcal{G}}}) \end{array}$$

That means that it suffices to show that

$$\gamma_D([x]_D) = \overline{M}_1 e(\gamma(x)) \qquad \text{for every } x \in A$$

is well-defined; indeed, since $\overline{M}_1 e \cdot \gamma$ is an M_0-algebra homomorphism and $\sim_{\mathcal{G}}$ a congruence, γ is then clearly an M_0-algebra homomorphism, too, and the above diagram commutes so that e becomes a coalgebra morphism that identifies all x, y such that D wins (x,y)).

So let D win (x,y); we have to show that

$$\overline{M}_1 e(\gamma(x)) = \overline{M}_1 e(\gamma(y)).$$

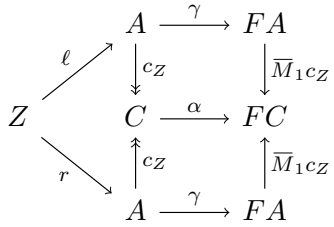
Let Z be the winning move of D at (x,y) (that is, Z is D's first move of the corresponding match). Since S can pick any element of Z, D wins on every element of Z, so that e factors through the above coequalizer $c_Z\colon A \to C_Z$: there is a unique M_0-algebra homomorphism $h\colon C_Z \to A/{\sim_{\mathcal{G}}}$ such that $h \cdot c_Z = e$. Hence, $\overline{M}_1 e$ factors through $\overline{M}_1 c_Z$ via $\overline{M}_1 h$. Since Z is an admissible move, $\overline{M}_1 c_Z$ identifies $\gamma(x)$ and $\gamma(y)$; hence, so does $\overline{M}_1 e$, as required.

($\Rightarrow$) Let $h\colon (A,\gamma) \to (B,\delta)$ be an $\overline{M}_1$-coalgebra morphism; it suffices to show that the kernel

$$\ker h = \{(x,y) \in A \times A \mid h(x) = h(y)\}$$

of h is contained in $\sim_{\mathcal{G}}$. To see this, it suffices to show that D can maintain the invariant $\ker h$, i.e. ensure that her positions always remain in $\ker h$ if the game starts in a position in $\ker h$. But when at a position $(x,y) \in \ker h$, she can clearly ensure that the next position is still in $\ker h$ by just playing $Z := \ker h$.

We proceed to show that Z is admissible. Let $\ell, r\colon Z \to A$ be the obvious projection maps. Since the kernel is clearly a congruence, Z is an M_0-algebra, and $\ell, r\colon Z \to A$ are M_0-algebra homomorphims. (In fact, Z is the pullback of h along itself, and the forgetful functor $\mathsf{Alg}_0(\mathbb{M}) \to \mathbf{Set}$ creates limits.) Using Remark 6.2.(1) (and noting that it holds for arbitrary algebras A in lieu of M_0X) we take the coequalizer $c_Z\colon A \twoheadrightarrow C_Z$ of the pair ℓ, r, and we now prove that $\overline{M}_1 c_Z \cdot \gamma$ merges the pair ℓ, r. By Remark A.3, the unique morphism $m\colon C_Z \rightarrowtail B$ such that $h = m \cdot c_Z$ is monic. By Remark A.8 we know that C_Z carries a unique coalgebra structure $\alpha\colon C_Z \to \overline{M}_1 C_Z$ such that c_Z and m are coalgebra morphisms. Using the former fact and that c_Z merges ℓ and r we obtain the following commutative diagram:



The commutativity of its outside is the desired equality. □

Proof of Theorem 7.12. It is straightforward to see that $\sim_{\mathcal{G}}$ is an equivalence relation. For instance, transitivity is seen as follows. Assume that D wins on (s,t) and on (t,u), with winning (first) moves Z, Z', respectively. Then $Z \cup Z'$ is a winning move for D on (s,u), where we exploit that by the assumption that $\overline{M}_1$ preserves monos, we do not need to worry about $Z \cup Z'$ possibly using more variables than needed in the top-level decompositions of s and u (cf. Remark 6.8). Symmetry and reflexivity are easier. We write $[t]^D = q(t)$. The M_0-algebra structure of U is then given by

$$f^U((\![t_i]\!]^D)_{i\in I}) = [\![f((t_i)_{i\in I})]\!]^D$$

for a depth-0 operation f of arity I. Well-definedness is seen similarly as transitivity above.

For well-definedness of ζ, suppose that $t_1\sigma, s_1\sigma$ are top-level decompositions of infinite-depth terms such that D wins on $(t_1\sigma, s_1\sigma)$, again w.l.o.g. with $t_1, s_1 \in \mathscr{T}_{\Sigma,1}(X)$ for the same X, and with the same σ. Let $Z \subseteq \mathscr{T}_{\Sigma,1}(X) \times \mathscr{T}_{\Sigma,1}(X)$ be D's winning move. Then Z, being an admissible move of D, entails $t_1 = s_1$. Moreover, since S can pick any element $(u,v) \in Z$ as a response, D wins on each of the subsequent positions $(u\sigma, v\sigma)$, so u and v are identified under $q \cdot \sigma$. It follows that $M_1(q\cdot\sigma)([t_1]) = M_1(q\cdot\sigma)([s_1])$. This shows well-definedness; it is then clear by construction that ζ is an M_0-homomorphism.

Finally, let $\gamma\colon A \to \overline{M}_1 A$ be an $\overline{M}_1$-coalgebra on an M_0-coalgebra A; we abuse A to denote also the carrier of A. For each $x \in A$, the successor structure $\gamma(x) \in \overline{M}_1 A$ is represented as a term $g_x \in \mathscr{T}_{\Sigma,1}(A)$. Infinite unrolling of A thus produces an infinite-depth term h_x for each $x \in A$. We claim that the map $h\colon A \to U$ given by $h(x) = [\![h_x]\!]^D$ is the unique $\overline{M}_1$-coalgebra morphism from (A,γ) to (U,ζ).

First note that the map h is independent of the choice of representing terms g_x. This is seen by letting D play, in $\mathcal{G}_\infty^{\mathsf{syn}}$, a strategy based on maintaining the invariant that the present state is a pair of unrollings, under differently chosen representing terms (where the choice of representing terms may change during the unrolling process), of the same state in A. Like in the 'only if' direction of the proof of Theorem 7.5, D can maintain the invariant by playing it in every move. To see that h is an M_0-homomorphism, let f be a depth-0 operation of arity I; then $\gamma(f(x_i)_{i\in I}) = f(\gamma(x_i)_{i\in I})$ has $f((g_{x_i})_{i\in I})$ as a representing term, so the unrolling $h(f(x_i)_{i\in I})$ arises by applying f to the unrollings $h(x_i)$.

Finally, being a coalgebra morphism $A \to U$ amounts precisely to the unrolling property, so h is a coalgebra morphism, and unique as such. □

Proof of Proposition 8.5.

Proof. We first show that Z is admissible at (s,t), i.e. $Z \vdash (\alpha_X \cdot \gamma)_0^* s = (\alpha_X \cdot \gamma)_0^* t$, where we write $Z \vdash$ to indicate equational derivability from Z. This follows from the construction of $\bar{Z}$ since Condition 1 ensures that

$$Z \vdash s_a + t_a = \sum_{x \in s_a} x + t_a = t_a.$$

Likewise Condition 2 ensures that $Z \vdash s_a + t_a = s_a$. Thus, $Z \vdash s_a = t_a$; whence, $Z \vdash (\alpha_X \cdot \gamma)_0^* s = (\alpha_X \cdot \gamma)_0^* t$.

Moreover, Spoiler thanks to Condition 1 (resp. Condition 2) can play the positions either of the form $(x \vee t', t')$ or $(y \vee s', s')$. In either cases, $M_0!$ will map the left and right term to 1 since $M_0 1 = 1$. So, in hindsight, if $(s,t)Z(s',t')$ is a winning match for Duplicator, then $s'\ Z\ t'$. Thus, repeating the above argument n-times give the desired result. Lastly, Z is also a winning strategy in the infinite game because $(s',t') \in Z$ whenever $(s,t)Z(s',t')$ is a match. □

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